ANSWER KEY Reduced copy of the question page, with its answer

Write the coefficient matrix of the system $2x + y = 7$ and $x - 3y = 0$. Give its rows.

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

801

QUESTION ID
S26-F1-Q1

TYPE
Short answer

CORRECT ANSWER
[2 1] and [1 −3]

GRADING NOTE Form required: the two rows in order. Each row lists the coefficients of x and y in one equation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q1/5

An augmented matrix has rows [1 2 | 5] and [3 −1 | 1].

Write the system it represents.

System: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

802

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

x + 2y = 5 and
3x − y = 1

GRADING NOTE Any equivalent form. Each row is one equation; the entry after the bar is the constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q2/5

On the rows [1 2 | 5] and [3 −1 | 1], replace row 2 by (row 2) − 3 × (row 1).

What is the new row 2?

A. [0 −7 | 14]

B. [2 −3 | −4]

C. [0 −7 | −14]

D. [0 5 | 16]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

803

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. [0 −7 | −14]

GRADING NOTE 3 − 3 = 0, −1 − 6 = −7, and 1 − 15 = −14.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q3**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q3/5

True or false?

A row operation on an augmented matrix does the same thing as replacing an equation by its sum with a multiple of another equation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

804

QUESTION ID
S26-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Each row is an equation, so the row operation is the elimination step written in matrix form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q4/5

True or false?

The coefficient matrix of $x + y = 3$ and $2x - y = 0$ has the rows [1 1] and [2 1].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

805

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The second row is [2 −1].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.8 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-REI.C.5

FORWARD 2

Use determinants and inverse matrices • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10

Find the determinant of the matrix with rows [3 1] and [5 2].

Determinant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

807

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form. 3(2) − 1(5) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q1/5

Show that the matrix with rows [2 −1] and [−5 3] is the inverse of the matrix with rows [3 1] and [5 2].

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

808

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The product has rows [3(2) + 1(−5) 3(−1) + 1(3)] = [1 0] and [5(2) + 2(−5) 5(−1) + 2(3)] = [0 1], the identity matrix.

GRADING NOTE Any equivalent work that multiplies the matrices and gets the identity matrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q2/5

Which matrix has no inverse?

A. Rows [2 4] and [1 2]

B. Rows [1 0] and [0 1]

C. Rows [2 1] and [1 1]

D. Rows [3 1] and [5 2]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

809

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [2 4] and [1 2]

GRADING NOTE 2(2) − 4(1) = 0. A 2-by-2 matrix has an inverse only when its determinant is not 0; the others have determinant 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system of two linear equations has an inverse, the system has exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

810

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The solution is $X = A^{-1}B$, and it is the only one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q4/5

True or false?

The inverse of the matrix with rows [2 0] and [0 4] has rows [−2 0] and [0 −4].

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

811

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The inverse has rows $[\frac{1}{2}\ 0]$ and $[0\ \frac{1}{4}]$: each diagonal entry is replaced by its reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.10 • Set 26 • FORWARD 2 • Q5/5

Solve systems of linear equations exactly by substitution and elimination

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Eliminate a variable by adding a multiple of an equation	**HSA-REI.C.5**
BACKWARD 2	Solve simple systems of linear equations algebraically	**8.EE.C.8.b**
FORWARD 1	Use slopes and systems to find where parallel or perpendicular li..	**HSG-GPE.B.5**
FORWARD 2	Multiply a matrix by a column to check a solution	**HSN-VM.C.8**

SET 27 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

Solve by substitution.

$y = 3x - 5$

$2x + y = 15$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

814

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 7)

GRADING NOTE Any equivalent form, such as x = 4, y = 7. 2x + 3x − 5 = 15, so x = 4 and y = 7. Check: 8 + 7 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q1/5

Solve by elimination.

$3x + 4y = 10$

$5x - 2y = 34$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

815

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −2)

GRADING NOTE Any equivalent form. Multiply the second equation by 2 and add: 13x = 78, so x = 6 and 18 + 4y = 10 gives y = −2. Check: 30 + 4 = 34.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q2/5

Solve.

$\frac{1}{2}x + \frac{1}{3}y = 4$

$x - y = 3$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

816

QUESTION ID

S27-M-Q3

TYPE

Short answer

CORRECT ANSWER

(6, 3)

GRADING NOTE Any equivalent form. Multiply the first equation by 6: 3x + 2y = 24. Substitute x = y + 3: 5y + 9 = 24, so y = 3. Check: 3 + 1 = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q3/5

Solve.

$2x - 3y = -1$

$4x + y = 19$

A. $(-4, -3)$

B. $(4, 3)$

C. $(3, 4)$

D. $(5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

817

QUESTION ID

S27-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (4, 3)

GRADING NOTE $y = 19 - 4x$, so $2x - 57 + 12x = -1$ and $x = 4$, $y = 3$. Check: $8 - 9 = -1$ and $16 + 3 = 19$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q4/5

True or false?

The system 2x + y = 5 and 4x + 2y = 9 has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

818

QUESTION ID
S27-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Doubling the first equation gives 4x + 2y = 10, which cannot also equal 9. The lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 27 • MAIN • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 1

Eliminate a variable by adding a multiple of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5

Add the equations x + 2y = 9 and 3x − 2y = 3.

What equation results, and what is x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

820

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4x = 12; x = 3

GRADING NOTE Form required: the new equation and x. The y-terms cancel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q1/5

System: 2x + y = 8 and 6x − 5y = 8.

What multiple of the first equation should you subtract from the second to eliminate x?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

821

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6x − 3(2x) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q2/5

System: x + y = 5 and 2x + 3y = 12.

Which equation can replace the second equation without changing the solution?

A. x = 2

B. y = 7

C. y = 2

D. 3x + 4y = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

822

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. y = 2

GRADING NOTE (2x + 3y) − 2(x + y) = 12 − 10 gives y = 2. The solution is (3, 2), which does not satisfy the other choices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q3/5

True or false?

Multiplying one equation of a system by a nonzero number does not change the solutions of the system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

823

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The new equation has exactly the same solutions as the old one.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q4/5

True or false?

Adding the equations $x + y = 4$ and $x - y = 2$ gives $2x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

824

QUESTION ID
S27-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $4 + 2 = 6$, so $2x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.5 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

BACKWARD 2

Solve simple systems of linear equations algebraically • Grade 8 • Expressions & Equations • 8.EE.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = x + 2$

$y = 3x$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

826

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 3)

GRADING NOTE Any equivalent form. $3x = x + 2$, so $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 10$

$y = 4$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

827

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(6, 4)

GRADING NOTE Any equivalent form. $x + 4 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q2/5

Solve.

$y = 2x$

$x + y = 12$

A. (4, 8)

B. (6, 6)

C. (8, 4)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

828

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (4, 8)

GRADING NOTE $x + 2x = 12$, so $x = 4$ and $y = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q3/5

True or false?

The system $y = 2x + 3$ and $y = 2x - 1$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

829

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Same slope, different y-intercepts: the lines are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q4/5

True or false?

The solution of $y = x$ and $y = -x + 6$ is $(2, 4)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

830

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x = -x + 6$ gives $x = 3$, so the solution is $(3, 3)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.b • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 1

Use slopes and systems to find where parallel or perpendicular lines meet • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Line ℓ is $y = 2x + 1$. Write an equation of the line through $(5, 1)$ that is perpendicular to ℓ, and find where the two lines meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

832

QUESTION ID
S27-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$y = -\frac{1}{2}x + 3.5$; the lines meet at (1, 3).

GRADING NOTE Any equivalent equation; the point is required. The slope is $-\frac{1}{2}$. $2x + 1 = -\frac{1}{2}x + 3.5$ gives $x = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Are the lines $2x + 3y = 6$ and $3x - 2y = 4$ perpendicular? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

833

QUESTION ID
S27-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes. Their slopes are $-\frac{2}{3}$ and $\frac{3}{2}$, and the product is -1.

GRADING NOTE Any equivalent explanation using slopes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q2/5

Which line is parallel to y = 3x − 2 and passes through (0, 5)?

A. y = 5x + 3

B. y = −3x + 5

C. y = 3x + 5

D. $y = -\frac{1}{3}x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

834

QUESTION ID
S27-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. y = 3x + 5

GRADING NOTE Parallel lines have the same slope, 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q3/5

True or false?

The lines $y = 4x + 1$ and $y = 4x - 7$ have no point in common.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

835

QUESTION ID
S27-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE They are parallel and distinct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q4/5

True or false?

The lines y = 2x and y = −2x are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

836

QUESTION ID
S27-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 2(−2) = −4, not −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.5 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-REI.C.6

FORWARD 2

Multiply a matrix by a column to check a solution • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8

Multiply the matrix with rows [2 1] and [1 −3] by the column [4, 1].

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

838

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The column [9, 1]

GRADING NOTE Any equivalent form. 2(4) + 1(1) = 9 and 1(4) − 3(1) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q1/5

The matrix A has rows [2 1] and [1 −3].

Use multiplication by A to decide whether x = 4, y = 1 solves 2x + y = 9 and x − 3y = 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

839

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. A times the column [4, 1] is the column [9, 1], which matches the constants.

GRADING NOTE Any equivalent explanation using the product.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q2/5

What is the product of the matrix with rows [1 2] and [3 4] and the column [1, 1]?

A. The column [3, 7]

B. The column [2, 6]

C. The column [10]

D. The column [4, 6]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

840

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The column [3, 7]

GRADING NOTE 1 + 2 = 3 and 3 + 4 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F2-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q3/5

True or false?

Multiplying a 2-by-2 matrix by a 2-by-1 column gives a 2-by-1 column.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

841

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(2 \times 2)(2 \times 1)$ gives 2×1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q4/5

True or false?

For all 2-by-2 matrices A and B, AB = BA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

842

QUESTION ID

S27-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Matrix multiplication is not commutative in general.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.8 • Set 27 • FORWARD 2 • Q5/5

Solve systems of linear equations in context, including systems with no solution or infinitely many solutions

MAIN	Algebra I main question	**HSA-REI.C.6**
BACKWARD 1	Write and solve an equation for one quantity in context	**HSA-CED.A.1**
BACKWARD 2	Solve real-world problems with two linear equations	**8.EE.C.8.c**
FORWARD 1	Use matrices to organize data in context	**HSN-VM.C.6**
FORWARD 2	Solve a system in context with an inverse matrix	**HSA-REI.C.9**

SET 28 • HSA-REI.C.6

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

A theater sold 340 tickets. Adult tickets cost $12 and student tickets cost $7. Sales totaled $3,255.

Write a system and find how many of each kind were sold.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

845

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

a + s = 340 and 12a + 7s = 3,255; 175 adult and 165 student tickets

GRADING NOTE Form required: a system and both numbers. 12a + 7(340 − a) = 3,255 gives 5a = 875. Check: 2,100 + 1,155 = 3,255.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q1/5

How many solutions does the system $6x - 4y = 10$ and $3x - 2y = 5$ have? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

846

QUESTION ID
S28-M-Q2

TYPE
Short answer

CORRECT ANSWER
Infinitely many. The first equation is 2 times the second, so both describe the same line.

GRADING NOTE Form required: the number of solutions and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q2/5

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution.

How many liters of each are needed?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

847

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

5 liters of the 10% solution and 15 liters of the 30% solution

GRADING NOTE Form required: both amounts, labeled. $x + y = 20$ and $0.10x + 0.30y = 5$. Then $6 - 0.2x = 5$, so $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q3/5

For which value of k does the system y = 3x + 2 and 6x − 2y = k have infinitely many solutions?

A. k = 2

B. k = −2

C. k = 4

D. k = −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

848

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. k = −4

GRADING NOTE Substitute: 6x − 2(3x + 2) = −4 for every x, so the equations describe the same line only when k = −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q4/5

True or false?

A system of two linear equations in two variables can have exactly two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

849

QUESTION ID
S28-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Two lines meet at no point, one point, or every point (the same line).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 28 • MAIN • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 1

Write and solve an equation for one quantity in context • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

A gym has 340 members. There are 40 more adults than teens.

Write and solve an equation for the number of teens t.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

851

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

t + (t + 40) = 340;
t = 150

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pens cost $2 and notebooks cost $5. Ana buys 3 notebooks and some pens and spends $27.

Write and solve an equation for the number of pens p.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

852

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2p + 15 = 27; p = 6

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q2/5

A 20-liter mix has x liters of a 10% acid solution and the rest a 30% acid solution.

Which expression gives the liters of acid?

A. 0.40(20)

B. 0.10x + 0.30x

C. 0.10x + 0.30(20 − x)

D. 10x + 30(20 − x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

853

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C.

0.10x + 0.30(20 − x)

GRADING NOTE The rest of the mix is 20 − x liters; 10% = 0.10 and 30% = 0.30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q3/5

True or false?

25 coins are all dimes and quarters. If d coins are dimes, then 25 − d coins are quarters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

854

QUESTION ID
S28-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The two counts add to 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

"3 more than 5 times a number is 48" gives 5n + 3 = 48, and its solution is n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

855

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 5n = 45, so n = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

BACKWARD 2

Solve real-world problems with two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

Two numbers have a sum of 30 and a difference of 8.

Find the numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

857

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

19 and 11

GRADING NOTE Any order. Adding the equations gives 2x = 38.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q1/5

Plan A costs $10 plus $2 per hour. Plan B costs $4 per hour.

After how many hours do they cost the same?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

858

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. 10 + 2h = 4h, so h = 5 (both cost $20).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

3 pencils and 2 erasers cost $1.30. 1 pencil and 2 erasers cost $0.70.

What does one pencil cost?

A. $0.60

B. $0.20

C. $0.35

D. $0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

859

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $0.30

GRADING NOTE Subtracting the equations gives 2 pencils = $0.60.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q3/5

True or false?

Two cars leave the same place at the same time in the same direction, at 50 mi/h and 60 mi/h. After 2 hours the faster car is 20 miles ahead.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

860

QUESTION ID
S28-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $120 - 100 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q4/5

True or false?

One company charges y = 2x + 5 dollars and another charges y = 2x + 8 dollars for x units. They charge the same for 3 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

861

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The lines are parallel, so the costs are never equal; the second always costs $3 more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 1

Use matrices to organize data in context • High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6

Friday: 120 adult and 90 student tickets. Saturday: 150 adult and 60 student tickets.

Write a 2-by-2 matrix with one row per day (adult first).

Rows: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

863

QUESTION ID
S28-F1-Q1

TYPE
Short answer

CORRECT ANSWER
[120 90] and [150 60]

GRADING NOTE Form required: Friday row first, adult column first.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q1**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q1/5

Adult tickets cost $12 and student tickets $7.

Multiply the matrix with rows [120 90] and [150 60] by the column [12, 7]. What do the entries mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The column [2,070, 2,220]: Friday sales were $2,070 and Saturday sales were $2,220.

GRADING NOTE Form required: both entries and their meaning. 1,440 + 630 = 2,070 and 1,800 + 420 = 2,220.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q2/5

A matrix has rows [5 2] and [3 4]. Each row is a store and each column is a product.

What does the entry in row 2, column 1 represent?

A. Store 2's amount of product 1

B. Store 1's amount of product 2

C. Store 2's amount of product 2

D. The total of product 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

865

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Store 2's amount of product 1

GRADING NOTE Row 2 is store 2, and column 1 is product 1. The entry is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q3**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q3/5

True or false?

A matrix with 3 rows and 2 columns is a 3-by-2 matrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

866

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Dimensions are written rows by columns.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q4/5

True or false?

Any two matrices can be added, whatever their dimensions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

867

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only matrices with the same dimensions can be added.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: Vector & Matrix Quantities • HSN-VM.C.6 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-REI.C.6

FORWARD 2

Solve a system in context with an inverse matrix • High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Write the system a + s = 340 and 12a + 7s = 3,255 as a matrix equation AX = B.

Give the rows of A.

Rows of A: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

869

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

[1 1] and [12 7]

GRADING NOTE Form required: the rows in order. X is the column [a, s] and B is the column [340, 3,255].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q1**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q1/5

The matrix with rows [1 1] and [12 7] has determinant −5.

Does the ticket system have exactly one solution? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

870

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The determinant is not 0, so A has an inverse and $X = A^{-1}B$ is the only solution.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F2-Q2**

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q2/5

The inverse of the matrix with rows [1 1] and [12 7] is (1 ÷ (−5)) times which matrix?

A. Rows [7 −1] and [−12 1]

B. Rows [7 1] and [12 1]

C. Rows [−7 1] and [12 −1]

D. Rows [1 −1] and [−12 7]

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

871

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Rows [7 −1] and [−12 1]

GRADING NOTE For rows [a b] and [c d], the inverse is (1 ÷ (ad − bc)) times rows [d −b] and [−c a].

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q3/5

True or false?

If the coefficient matrix of a system has determinant 0, the system does not have exactly one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

872

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The matrix has no inverse; the system has no solution or infinitely many.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q4/5

True or false?

The determinant of the matrix with rows [2 3] and [4 6] is 24.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

873

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2(6) - 3(4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School — Precalculus (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.9 • Set 28 • FORWARD 2 • Q5/5

Solve a system of a linear equation and a quadratic equation algebraically

MAIN	Algebra I main question	**HSA-REI.C.7**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Solve systems of linear equations	**HSA-REI.C.6**
FORWARD 1	Decide whether points lie on a circle	**HSG-GPE.B.4**
FORWARD 2	Find complex solutions when a line and a parabola do not meet	**HSN-CN.C.7**

SET 29 • HSA-REI.C.7

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7

Solve the system.

$y = x^2 - 3$

$y = 2x$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

876

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6) and (−1, −2)

GRADING NOTE Form required: both ordered pairs. $x^2 - 3 = 2x$ gives $x^2 - 2x - 3 = (x - 3)(x + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q1/5

Solve the system.

$y = x^2 + 4x + 1$

$y = x - 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

877

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−1, −2) and (−2, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + 3x + 2 = (x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q2/5

How many real solutions does the system $y = x^2 + 2$ and $y = x$ have?

A. Infinitely many

B. One

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

878

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE $x^2 - x + 2 = 0$ has discriminant $1 - 8 = -7 < 0$, so the line and the parabola do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q3/5

Solve the system.

$x^2 + y^2 = 25$

$y = x + 1$

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

879

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

(3, 4) and (−4, −3)

GRADING NOTE Form required: both ordered pairs. $x^2 + (x + 1)^2 = 25$ gives $x^2 + x - 12 = 0$, so $x = 3$ or $x = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q4/5

True or false?

The line $y = 2x - 1$ meets the parabola $y = x^2$ at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

880

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 - 2x + 1 = (x - 1)^2 = 0$ has one solution, $x = 1$. The line touches the parabola at (1, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.7 • Set 29 • MAIN • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 - 2x - 3 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

882

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −1

GRADING NOTE Form required: both solutions. $(x - 3)(x + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + x - 12 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

883

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

x = 3 or x = −4

GRADING NOTE Form required: both solutions. (x + 4)(x − 3) = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 3x + 2 = 0$.

A. $x = -1$ or $x = 2$

B. $x = 1$ or $x = 2$

C. $x = 1$ or $x = -2$

D. $x = -1$ or $x = -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

884

QUESTION ID

S29-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = -1$ or $x = -2$

GRADING NOTE $(x + 1)(x + 2) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q3/5

True or false?

$x^2 - x + 2 = 0$ has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

885

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The discriminant is $1 - 8 = -7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q4/5

True or false?

The solutions of $x^2 = 2x$ are $x = 2$ and $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

886

QUESTION ID
S29-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $x(x - 2) = 0$, so $x = 0$ or $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

BACKWARD 2

Solve systems of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$y = 2x$

$y = -x + 9$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

888

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6)

GRADING NOTE Any equivalent form. $2x = -x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 1$

$y = x + 1$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

889

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. $x + x + 1 = 1$, so $x = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 7$

$x - y = 2$

A. (4, −1)

B. (3, 1)

C. (1, 3)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

890

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 1)

GRADING NOTE Adding gives 3x = 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q3/5

True or false?

Substitution and elimination give the same solution of a system.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

891

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both methods keep the solutions unchanged at every step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q4/5

True or false?

(2, 1) is a solution of the system $x + y = 3$ and $x - y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

892

QUESTION ID
S29-B2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 - 1 = 1$, not 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 1

Decide whether points lie on a circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

A circle is centered at the origin and passes through (0, 5).

Does the point (3, −4) lie on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

894

QUESTION ID
S29-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. The radius is 5, and the distance from (0, 0) to (3, −4) is √(9 + 16) = 5.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the points where the line $y = 3$ meets the circle $x^2 + y^2 = 25$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

895

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 3) and (−4, 3)

GRADING NOTE Form required: both points. $x^2 + 9 = 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q2/5

A circle has center (1, 2) and radius 5.

Which point is on the circle?

A. (1, 6)

B. (4, 6)

C. (3, 3)

D. (6, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

896

QUESTION ID
S29-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. (4, 6)

GRADING NOTE From (1, 2) to (4, 6): √(9 + 16) = 5. The others are √5, √26, and 4 units away.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q3/5

True or false?

The line $y = 6$ does not meet the circle $x^2 + y^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

897

QUESTION ID
S29-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The circle reaches only up to $y = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q4/5

True or false?

The point (2, 2) lies on the circle $x^2 + y^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

898

QUESTION ID
S29-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 4 + 4 = 8, not 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.4 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-REI.C.7

FORWARD 2

Find complex solutions when a line and a parabola do not meet • High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7

ANSWER KEY Reduced copy of the question page, with its answer

Substituting $y = 2x$ into $y = x^2 + 2$ gives $x^2 - 2x + 2 = 0$.

Solve it over the complex numbers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

900

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 1 + i$ or $x = 1 - i$

GRADING NOTE Form required: a + bi (both solutions). $x = (2 \pm \sqrt{(-4)}) \div 2$. The solutions are not real, so the graphs do not meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x^2 + 4 = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

901

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 2i$ or $x = -2i$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q2**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q2/5

Solve $x^2 - 6x + 10 = 0$.

A. $3 + i$ and $3 - i$

B. $6 + i$ and $6 - i$

C. $3 + 2i$ and $3 - 2i$

D. $-3 + i$ and $-3 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

902

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3 + i and 3 − i

GRADING NOTE $(x - 3)^2 = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q3**

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q3/5

True or false?

When a line and a parabola do not intersect, the equation from substitution has no real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

903

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A real solution x would give a common point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q4/5

True or false?

$i^3 = i$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

904

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $i^3 = i^2 \cdot i = -i$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School — Algebra II (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 29 • FORWARD 2 • Q5/5

Understand the graph of an equation as the set of all its solutions

MAIN	Algebra I main question	**HSA-REI.D.10**
BACKWARD 1	Use tables and graphs to relate two variables	**6.EE.C.9**
BACKWARD 2	Understand a function's graph as its input-output pairs	**8.F.A.1**
FORWARD 1	Derive a parabola from its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use the unit circle as the set of points (cos θ, sin θ)	**HSF-TF.A.2**

SET 30 • HSA-REI.D.10

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (3, −2) on the graph of 2x + 5y = −4? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

907

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 2(3) + 5(−2) = 6 − 10 = −4, so (3, −2) is a solution.

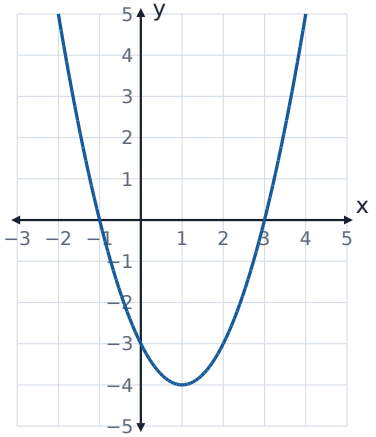
GRADING NOTE Any equivalent explanation that substitutes the point.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q1/5

The graph of $y = x^2 - 2x - 3$ is shown.

Which point is NOT a solution of the equation?



A. (2, 3) **B.** (3, 0)

C. (0, −3) **D.** (1, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

908

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 3)

GRADING NOTE At $x = 2$, $y = 4 - 4 - 3 = -3$, not 3. The other points are on the graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (k, 7) is on the graph of y = 3x − 2.

Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

909

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 3k − 2 = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q3/5

The graph of an equation in x and y passes through (4, 1) and (−2, 5).

What does that tell you about the equation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

910

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

x = 4, y = 1 and x = −2, y = 5 both make the equation true: each is a solution.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q4/5

True or false?

A point that is not on the graph of $y = x^2$ can still be a solution of $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

911

QUESTION ID
S30-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The graph is exactly the set of all solutions, so every solution is on it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 30 • MAIN • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 1

Use tables and graphs to relate two variables • Grade 6 • Expressions & Equations • 6.EE.C.9

ANSWER KEY Reduced copy of the question page, with its answer

y = 3x. Find y for x = 0, 1, 2, and 3.

Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

913

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0, 3, 6, 9

GRADING NOTE Any equivalent form, in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A car travels at 40 miles per hour.

Write an equation for the distance d after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

914

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

d = 40t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q2/5

Which point is on the graph of y = x + 4?

A. (2, 6)

B. (1, 3)

C. (4, 0)

D. (6, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

915

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 6)

GRADING NOTE 2 + 4 = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

In y = 5x, y is the dependent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

916

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE y depends on the value chosen for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (2, 8) is on the graph of y = 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

917

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3(2) = 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

BACKWARD 2

Understand a function's graph as its input-output pairs • Grade 8 • Functions • 8.F.A.1

ANSWER KEY Reduced copy of the question page, with its answer

The function f pairs each x with 2x − 1.

What output does it pair with x = 4?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

919

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is {(1, 2), (2, 3), (1, 5)} a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

920

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 1 is paired with two outputs, 2 and 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q2/5

Which point is on the graph of $y = x^2$?

A. (9, −3)

B. (−3, −9)

C. (−3, 9)

D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

921

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 9)

GRADING NOTE $(-3)^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

922

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

The vertical line x = 2 is the graph of a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

923

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The input 2 would have infinitely many outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 1

Derive a parabola from its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

A parabola has focus (0, 2) and directrix y = −2.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

925

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 = 8y$

GRADING NOTE Any equivalent form, such as $y = x^2 \div 8$. Setting $\sqrt{(x^2 + (y - 2)^2)} = |y + 2|$ and squaring gives $x^2 - 4y = 4y$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q1/5

Is (4, 2) on the parabola with focus (0, 2) and directrix $y = -2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

926

QUESTION ID
S30-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes. It is 4 units from the focus and 4 units from the directrix.

GRADING NOTE Any equivalent explanation (or $4^2 = 8 \cdot 2$).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q2/5

Where is the focus of the parabola $y = x^2 \div 4$?

A. (0, 4)

B. $(0, \frac{1}{4})$

C. (1, 0)

D. (0, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5

927

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 1)

GRADING NOTE $x^2 = 4y = 4py$ with $p = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F1-Q3**

**High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q3/5**

True or false?

Every point on a parabola is the same distance from its focus as from its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

928

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition of a parabola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q4/5

True or false?

The vertex of a parabola lies on its directrix.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

929

QUESTION ID
S30-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The vertex is halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-REI.D.10

FORWARD 2

Use the unit circle as the set of points (cos θ, sin θ) • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

Explain why the point $(\cos\theta, \sin\theta)$ lies on the circle $x^2 + y^2 = 1$ for every angle θ.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

931

QUESTION ID

S30-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$\cos^2\theta + \sin^2\theta = 1$, so the point satisfies the equation. (It is 1 unit from the origin.)

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point on the unit circle corresponds to θ = π ÷ 2?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

932

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 1)

GRADING NOTE Any equivalent form. cos(π ÷ 2) = 0 and sin(π ÷ 2) = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S30-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q2/5

Which point is on the unit circle $x^2 + y^2 = 1$?

A. (0.5, 0.5)

B. (1, 1)

C. (0.6, 0.8)

D. (0.8, 0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

933

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (0.6, 0.8)

GRADING NOTE 0.36 + 0.64 = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q3/5

True or false?

The point $(-1, 0)$ on the unit circle corresponds to an angle of π radians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

934

QUESTION ID
S30-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $\cos \pi = -1$ and $\sin \pi = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q4/5

True or false?

The point (cos 0, sin 0) is (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

935

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE cos 0 = 1 and sin 0 = 0, so the point is (1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 30 • FORWARD 2 • Q5/5

Explain why intersections of y = f(x) and y = g(x) solve f(x) = g(x), and approximate solutions with graphs and tables

MAIN	Algebra I main question	**HSA-REI.D.11**
BACKWARD 1	Check whether points are on a graph	**HSA-REI.D.10**
BACKWARD 2	Read the solution of a system from the graphs of two lines	**8.EE.C.8.a**
FORWARD 1	Solve exponential equations exactly with logarithms	**HSF-LE.A.4**
FORWARD 2	Solve trigonometric equations with graphs and technology	**HSF-TF.B.7**

SET 31 • HSA-REI.D.11

MAIN

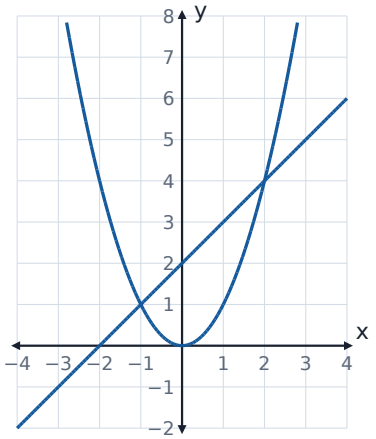
Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $f(x) = x^2$ and $g(x) = x + 2$ are shown.

Use the graphs to solve $x^2 = x + 2$.

Solutions: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

938

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = -1$ or $x = 2$

GRADING NOTE Form required: both solutions. The graphs meet at $(-1, 1)$ and $(2, 4)$. Check: $(-1)^2 = 1 = -1 + 2$ and $2^2 = 4 = 2 + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use a table of values to find the solution of $2^x = 3x$ between $x = 3$ and $x = 4$, to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

939

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.3

GRADING NOTE Form required: rounded to the nearest tenth. At 3.3, $2^x \approx 9.85 < 9.9 = 3x$; at 3.35, $2^x \approx 10.20 > 10.05 = 3x$. The solution is about 3.31.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q2/5

The table shows f(x) = |x − 1| and g(x) = 3 − x.

For which x is f(x) = g(x)?

x	0	1	2	3	4
f(x)	1	0	1	2	3
g(x)	3	2	1	0	−1

A. x = 2 **B.** x = 3 **C.** x = 0 **D.** x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

940

QUESTION ID

S31-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 2

GRADING NOTE At x = 2 both equal 1. In every other column the outputs differ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q3/5

Explain why the x-coordinates of the points where the graphs of y = f(x) and y = g(x) meet are the solutions of f(x) = g(x).

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

941

QUESTION ID

S31-M-Q4

TYPE

Short answer

CORRECT ANSWER

At a meeting point (a, b), b = f(a) and b = g(a), so f(a) = g(a). If f(a) = g(a), the point (a, f(a)) is on both graphs.

GRADING NOTE Any equivalent explanation that uses the coordinates of a point on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q4/5

True or false?

The graphs of $y = 3x + 1$ and $y = 3x - 4$ never meet, so $3x + 1 = 3x - 4$ has no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

942

QUESTION ID
S31-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The lines are parallel, and $1 = -4$ is false for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.D.11

BACKWARD 1

Check whether points are on a graph • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (2, 4) on both $y = x^2$ and $y = x + 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

944

QUESTION ID

S31-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $2^2 = 4$ and $2 + 2 = 4$.

GRADING NOTE Any equivalent explanation that checks both equations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point $(-1, y)$ is on the graph of $y = x + 2$.

Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

945

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = |x - 1|$?

A. $(-2, -3)$

B. $(-2, 3)$

C. $(3, -2)$

D. $(1, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

946

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, 3)

GRADING NOTE $|-2 - 1| = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q3/5

True or false?

If (a, b) is on the graph of y = f(x), then b = f(a).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

947

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is the set of points (x, f(x)).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q4/5

True or false?

(0, 1) is on the graph of $y = 2^x + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

948

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2^0 + 1 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

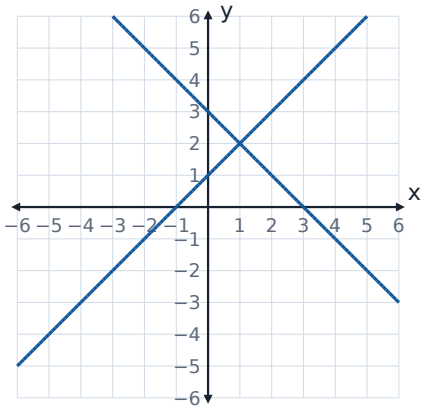
BACKWARD 2

Read the solution of a system from the graphs of two lines • Grade 8 • Expressions & Equations • 8.EE.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

Use the graph to solve the system $y = x + 1$ and $y = -x + 3$.

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

950

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(1, 2)

GRADING NOTE Any equivalent form. The lines cross at (1, 2). Check: 1 + 1 = 2 and −1 + 3 = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At what x-value do the graphs of $y = 2x - 3$ and $y = 5$ meet?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

951

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2x - 3 = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q2/5

The graphs of two linear equations are the same line.

How many solutions does the system have?

A. One

B. Infinitely many

C. Two

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

952

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Infinitely many

GRADING NOTE Every point of the line is on both graphs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q3/5

True or false?

Two lines with different slopes meet at exactly one point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

953

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lines with different slopes are not parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q4/5

True or false?

Two lines with the same slope always meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

954

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Distinct lines with the same slope are parallel and never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 1

Solve exponential equations exactly with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The graphs of $y = 2^x$ and $y = 10$ meet at one point.

Find its x-coordinate exactly using logarithms, and to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

956

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

x = log 10 ÷ log
2 = 1 ÷ log 2 ≈ 3.32

GRADING NOTE Form required: the exact expression (such as $\log_2 10$) and the rounded value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q1/5

Solve $5e^x = 20$ exactly.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

957

QUESTION ID

S31-F1-Q2

TYPE

Short answer

CORRECT ANSWER

ln 4

GRADING NOTE Form required: exact. $e^x = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q2/5

A graph shows $y = 3^x$ meeting $y = 15$ at about $x = 2.5$.

Which expression is the exact solution of $3^x = 15$?

A. 15 ÷ 3

B. log 5

C. log 15 ÷ log 3

D. √15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

958

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. log 15 ÷ log 3

GRADING NOTE Take the logarithm of both sides: x log 3 = log 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

$\log_2 8 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

959

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $2^3 = 8$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

The solution of $10^x = 0.01$ is $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

960

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $0.01 = 10^{-2}$, so $x = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S31-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.D.11

FORWARD 2

Solve trigonometric equations with graphs and technology • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Solve $\sin x = \frac{1}{2}$ for $0 \le x \le \pi$. Give exact answers.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

962

QUESTION ID

S31-F2-Q1

TYPE

Short answer

CORRECT ANSWER

x = π ÷ 6 or

x = 5π ÷ 6

GRADING NOTE Form required: exact, both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q1/5

Use a calculator to solve $\cos x = 0.3$ for $0 \le x \le \pi$, to the nearest hundredth of a radian.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

963

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.27

GRADING NOTE Form required: rounded to the nearest hundredth. $x = \cos^{-1}(0.3) \approx 1.266$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q2/5

How many solutions does sin x = 2 have?

A. Two

B. One

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

964

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. None

GRADING NOTE The values of sin x are between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S31-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graphs of y = sin x and y = 0.5 meet at infinitely many points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

965

QUESTION ID
S31-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The sine graph repeats every 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q4/5

True or false?

For $0 \le x \le \pi$, the equation $\cos x = 0$ has two solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

966

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only $x = \pi \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 31 • FORWARD 2 • Q5/5

Graph the solutions of linear inequalities and systems of linear inequalities in two variables as half-planes

MAIN	Algebra I main question	**HSA-REI.D.12**
BACKWARD 1	Solve an inequality in two variables for y	**HSA-REI.B.3**
BACKWARD 2	Plot points and find distances in the coordinate plane	**6.NS.C.8**
FORWARD 1	Find the area of a region described by inequalities	**HSG-GPE.B.7**
FORWARD 2	Use zeros to find where a polynomial is positive or negative	**HSA-APR.B.3**

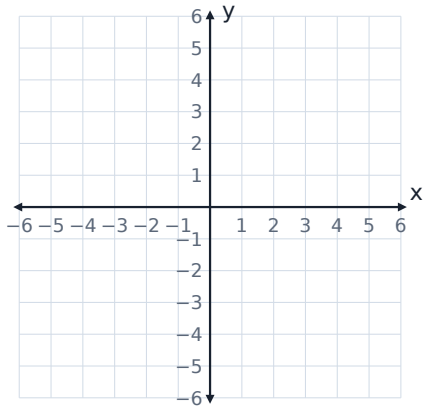
SET 32 • HSA-REI.D.12

MAIN

Algebra I main question • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Graph $y < 2x - 1$ on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

969

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A dashed line through (0, −1) and (2, 3), with the half-plane below and to the right of it shaded

GRADING NOTE Any correct graph. The line is dashed because points on it are not solutions; (0, −2) is a solution and (0, 0) is not.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q1/5

Is (4, 1) a solution of the system $y \leq -x + 6$ and $y > \frac{1}{2}x - 2$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

970

QUESTION ID

S32-M-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. $1 \leq -4 + 6 = 2$, and $1 > 2 - 2 = 0$.

GRADING NOTE Any equivalent explanation that checks both inequalities.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q2/5

A graph has a solid boundary line through (0, 2) and (3, 0), and the side that contains the origin is shaded.

Which inequality does it show?

A. $2x + 3y < 6$

B. $2x + 3y \geq 6$

C. $3x + 2y \leq 6$

D. $2x + 3y \leq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

971

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x + 3y \leq 6$

GRADING NOTE The line is $2x + 3y = 6$; solid means "or equal to"; (0, 0) gives $0 \leq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q3

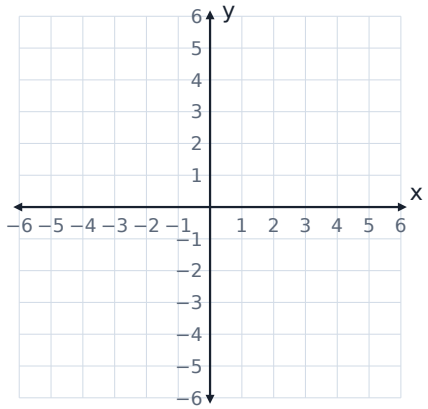
High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $y \geq x - 2$ and $x + y < 4$ on the coordinate plane.

Name one point that is a solution of the system.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

972

QUESTION ID

S32-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A solid line $y = x - 2$ with the side above it shaded, a dashed line $x + y = 4$ with the side below it shaded, and the overlap marked

Written: Any point in the overlap, such as (0, 0)

GRADING NOTE Any equivalent form: the point must satisfy both inequalities. (0, 0): $0 \geq -2$ and $0 < 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q4/5

True or false?

Points on the boundary line y = 3x are solutions of y > 3x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

973

QUESTION ID
S32-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE On the line, y = 3x, which is not greater than 3x. The boundary is dashed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 1

Solve an inequality in two variables for y • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 2x + 3y ≤ 6 for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

975

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$y \leq -\frac{2}{3}x + 2$

GRADING NOTE Any equivalent form with y alone. 3y ≤ −2x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $x - 2y > 4$ for y.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

976

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$y < \frac{1}{2}x - 2$

GRADING NOTE Any equivalent form with y alone. Dividing by −2 reverses the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-y \geq x - 5$ for y.

A. $y \geq x + 5$

B. $y \geq -x + 5$

C. $y \leq -x + 5$

D. $y \leq x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

977

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y \leq -x + 5$

GRADING NOTE Multiply by −1 and reverse the sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Solving $4x + y < 3$ for y gives $y < -4x + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

978

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Subtract 4x from both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q4/5

True or false?

Solving $-3y < 9$ for y gives $y < -3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

979

QUESTION ID
S32-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Dividing by −3 reverses the sign: $y > -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

BACKWARD 2

Plot points and find distances in the coordinate plane • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (2, −3) and (2, 5).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

981

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

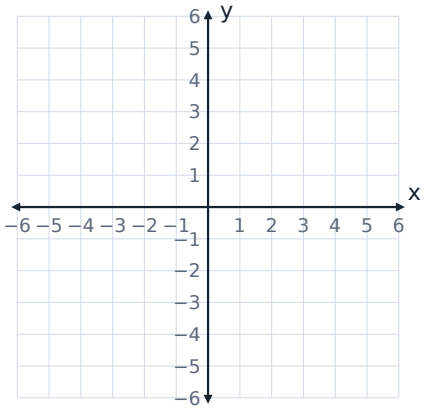
GRADING NOTE Any equivalent form. |5 − (−3)| = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot the points (−3, 2) and (4, −1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

982

QUESTION ID

S32-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A point at (−3, 2) and a point at (4, −1)

GRADING NOTE Any correct plot.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which point is in Quadrant III?

A. (2, −5)

B. (−2, −5)

C. (2, 5)

D. (−2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

983

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−2, −5)

GRADING NOTE Quadrant III has both coordinates negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

The points (3, 4) and (3, −4) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

984

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Only the sign of y changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The distance between (−4, 1) and (6, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

985

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $|6 - (-4)| = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 1

Find the area of a region described by inequalities • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

ANSWER KEY Reduced copy of the question page, with its answer

The system $x \geq 0$, $y \geq 0$, and $x + y \leq 6$ describes a triangle.

Find its vertices and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

987

QUESTION ID
S32-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(0, 0), (6, 0), and (0, 6); area 18 square units

GRADING NOTE Form required: the vertices and the area. $\frac{1}{2}(6)(6) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the perimeter of the rectangle with vertices (1, 1), (5, 1), (5, 4), and (1, 4).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

988

QUESTION ID
S32-F1-Q2

TYPE
Short answer

CORRECT ANSWER
14 units

GRADING NOTE Any equivalent form. 2(4 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q2/5

The region $0 \le x \le 4$ and $0 \le y \le 2x$ is a triangle.

What is its area?

A. 32

B. 16

C. 8

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

989

QUESTION ID
S32-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
B. 16

GRADING NOTE Vertices (0, 0), (4, 0), and (4, 8): $\frac{1}{2}(4)(8) = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q3/5

True or false?

The triangle with vertices (0, 0), (3, 0), and (0, 4) has perimeter 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

990

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3 + 4 + 5 = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q4/5

True or false?

The square with vertices (0, 0), (2, 0), (2, 2), and (0, 2) has area 8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

991

QUESTION ID
S32-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.7 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.D.12

FORWARD 2

Use zeros to find where a polynomial is positive or negative • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

ANSWER KEY Reduced copy of the question page, with its answer

$p(x) = (x - 1)(x - 4)$.

For which x is $p(x) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

993

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$1 < x < 4$

GRADING NOTE Form required: an inequality. Between the zeros, one factor is positive and one is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q1/5

$p(x) = x(x + 2)(x - 3)$.

Is $p(1)$ positive or negative? Explain using its factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

994

QUESTION ID
S32-F2-Q2

TYPE
Short answer

CORRECT ANSWER
Negative: $1 \cdot 3 \cdot (-2) = -6$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

For which x is $(x + 3)(x - 2) > 0$?

A. $x < -3$ only

B. $x > 2$ only

C. $x < -3$ or $x > 2$

D. $-3 < x < 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

995

QUESTION ID
S32-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $x < -3$ or $x > 2$

GRADING NOTE Both factors are positive when $x > 2$, and both are negative when $x < -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q3/5

True or false?

The graph of $y = (x - 1)^2$ touches the x-axis at $x = 1$ without crossing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

996

QUESTION ID
S32-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $(x - 1)^2$ is never negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)(x + 1) \geq 0$ for every real number x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

997

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0 it equals −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.B.3 • Set 32 • FORWARD 2 • Q5/5

Create equations in two or more variables and graph them on labeled axes

MAIN	Algebra I main question	**HSA-CED.A.2**
BACKWARD 1	Write and solve an equation in one variable	**HSA-CED.A.1**
BACKWARD 2	Write the equation of a linear function	**8.F.B.4**
FORWARD 1	Write the equation of a circle from a situation	**HSG-GPE.A.1**
FORWARD 2	Write and interpret polynomial models	**HSF-IF.C.7.c**

SET 33 • HSA-CED.A.2

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A savings account has $150 and grows by $25 each week.

Write an equation for the balance B after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

1000

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

B = 150 + 25w

GRADING NOTE Any equivalent form, such as B = 25w + 150.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q1

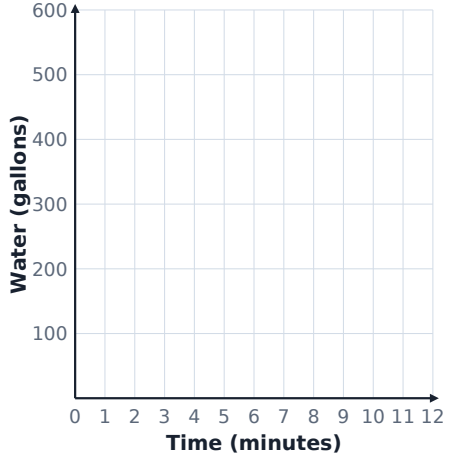
High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool holds 600 gallons and drains at 50 gallons per minute.

Write an equation for the gallons G left after t minutes, and graph it for 0 ≤ t ≤ 12.

Equation: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

1001

QUESTION ID

S33-M-Q2

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line from (0, 600) to (12, 0)

Written:

G = 600 − 50t

GRADING NOTE Any equivalent form. Both the equation and the graph are graded. The pool is empty at t = 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle's length is 3 more than twice its width w.

Write an equation for its area A.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

1002

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

A = w(2w + 3)

GRADING NOTE Any equivalent form, such as $A = 2w^2 + 3w$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A town has 8,000 people, and its population grows 3% each year.

Which equation gives the population P after t years?

A. $P = 8{,}000(0.03)^t$

B. $P = 8{,}000 + 0.03t$

C. $P = 8{,}000(1.03)^t$

D. $P = 8{,}000(1.3)^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

1003

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $P = 8{,}000(1.03)^t$

GRADING NOTE Each year the population is multiplied by 1.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q4/5

True or false?

Student tickets cost $4 and adult tickets cost $9. The equation 4s + 9a = 300 relates the numbers of tickets when sales total $300.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

1004

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 4s dollars from s student tickets plus 9a dollars from a adult tickets.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 33 • MAIN • Q5/5

SET 33 • HSA-CED.A.2

BACKWARD 1

Write and solve an equation in one variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Solve 150 + 25w = 400 for w.

w = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. 25w = 250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A pool has 600 gallons and drains at 50 gallons per minute.

Write and solve an equation for when it is empty.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

600 − 50t = 0; t = 12 minutes

GRADING NOTE Form required: an equation and its solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A plan costs $12 per month plus a $30 fee.

Which equation gives the number of months m for a total of $186?

A. 12m − 30 = 186

B. 30m + 12 = 186

C. 12(m + 30) = 186

D. 12m + 30 = 186

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12m + 30 = 186

GRADING NOTE 12 per month for m months, plus 30 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q3/5

True or false?

w = 2.5 is a solution of w(2w + 3) = 20.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 2.5(5 + 3) = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving 4s + 9(20) = 300 gives s = 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4s = 120, so s = 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 33 • BACKWARD 1 • Q5/5

1010

SET 33 • HSA-CED.A.2

BACKWARD 2

Write the equation of a linear function • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A line passes through (0, 5) and (2, 11).

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 3x + 5

GRADING NOTE Any equivalent form. The slope is 6 ÷ 2 = 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A plumber charges $50 plus $30 per hour.

Write a function for the cost C(h).

C(h) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30h + 50

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A table has x-values 0, 1, 2 and y-values 4, 7, 10.

Which equation fits the table?

A. y = 4x + 3

B. y = x + 4

C. y = 3x

D. y = 3x + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 3x + 4

GRADING NOTE y starts at 4 and increases by 3 for each 1 in x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q3/5

True or false?

The rate of change of $y = -2x + 7$ is -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The rate of change is the slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q4/5

True or false?

The initial value of $y = 6x - 1$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value (at $x = 0$) is -1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 1

Write the equation of a circle from a situation • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

A sprinkler at (3, −2) waters everything within 4 meters.

Write an equation of the boundary of the watered region in standard form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

1018

QUESTION ID
S33-F1-Q1

TYPE
Short answer

CORRECT ANSWER
$(x - 3)^2 + (y + 2)^2 = 16$

GRADING NOTE Form required: the standard form of a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q1**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A cell tower at the origin reaches 10 km. Is a phone at (6, 7) within range? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

1019

QUESTION ID
S33-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Yes.
$6^2 + 7^2 = 85 < 100$, so it is about 9.2 km away.

GRADING NOTE Any equivalent explanation using distance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation describes all the points 5 units from (0, 0)?

A. $x^2 - y^2 = 25$

B. $x^2 + y^2 = 25$

C. $x + y = 5$

D. $x^2 + y^2 = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5

1020

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x^2 + y^2 = 25$

GRADING NOTE The radius squared is 25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F1-Q3**

**High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q3/5**

True or false?

The circle $(x - 1)^2 + (y - 1)^2 = 4$ passes through (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

1021

QUESTION ID
S33-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $2^2 + 0^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The circle $x^2 + y^2 = 9$ passes through (3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

1022

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 9 + 9 = 18, not 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-CED.A.2

FORWARD 2

Write and interpret polynomial models • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

An open box is made from a 10 in by 8 in sheet by cutting a square of side x from each corner.

Write an equation for its volume V.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

1024

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

V = x(10 − 2x)(8 − 2x)

GRADING NOTE Any equivalent form. The base is (10 − 2x) by (8 − 2x), and the height is x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For V = x(10 − 2x)(8 − 2x), which values of x make sense in the situation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

1025

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0 < x < 4

GRADING NOTE Form required: an inequality. Each side length must be positive, and 8 − 2x > 0 means x < 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What are the zeros of $V = x(10 - 2x)(8 - 2x)$?

A. 5 and 4 only

B. 0, 10, and 8

C. 0, −5, and −4

D. 0, 5, and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

1026

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0, 5, and 4

GRADING NOTE Set each factor equal to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q3/5

True or false?

The graph of $y = x^3 - x$ crosses the x-axis three times.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

1027

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x(x - 1)(x + 1) = 0$ at $x = -1$, 0, and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q4/5

True or false?

As x gets very large, $y = -x^3$ becomes very large and positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

1028

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-x^3$ becomes very large and negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 33 • FORWARD 2 • Q5/5

Represent constraints with equations and inequalities and decide whether solutions are viable

MAIN	Algebra I main question	**HSA-CED.A.3**
BACKWARD 1	Test points in a system of inequalities	**HSA-REI.D.12**
BACKWARD 2	Solve a system of linear equations	**HSA-REI.C.6**
FORWARD 1	Use constraints to optimize a design	**HSG-MG.A.3**
FORWARD 2	Use expected values to compare decisions	**HSS-MD.B.7**

SET 34 • HSA-CED.A.3

MAIN

Algebra I main question • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A bakery makes m muffins and b loaves. A muffin uses 1 cup of flour and a loaf uses 3 cups; there are 60 cups. A muffin takes 2 minutes and a loaf takes 5 minutes; there are 140 minutes.

Write the system of constraints.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

1031

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

m + 3b ≤ 60, 2m + 5b ≤ 140, m ≥ 0, b ≥ 0 (m and b whole numbers)

GRADING NOTE Form required: both resource inequalities; the conditions m ≥ 0 and b ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For the bakery, is making 40 muffins and 10 loaves possible? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

1032

QUESTION ID
S34-M-Q2

TYPE
Short answer

CORRECT ANSWER
No. It needs 40 + 30 = 70 cups of flour, more than 60 (the 130 minutes would be enough).

GRADING NOTE Any equivalent explanation that checks the constraints.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q2/5

A club needs at least $500 from selling shirts at $15 and hats at $10, and it can sell at most 45 items.

Write the constraints for s shirts and h hats.

Constraints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

1033

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

15s + 10h ≥ 500, s + h ≤ 45, s ≥ 0, h ≥ 0 (whole numbers)

GRADING NOTE Form required: both inequalities; s, h ≥ 0 (whole numbers) are also expected.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

For the club, which combination meets both constraints?

A. 15 shirts and 25 hats

B. 25 shirts and 15 hats

C. 30 shirts and 20 hats

D. 20 shirts and 10 hats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

1034

QUESTION ID

S34-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 25 shirts and 15 hats

GRADING NOTE 375 + 150 = 525 ≥ 500 and 40 ≤ 45. The others give 50 items, $475, and $400.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q4/5

True or false?

If x counts people, x = 12.5 can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

1035

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A number of people must be a whole number, so 12.5 is not viable.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 34 • MAIN • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 1

Test points in a system of inequalities • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12

ANSWER KEY Reduced copy of the question page, with its answer

Is (10, 10) a solution of $x + 3y \leq 60$ and $2x + 5y \leq 140$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

1037

QUESTION ID
S34-B1-Q1

TYPE
Short answer

CORRECT ANSWER
Yes. 40 ≤ 60 and 70 ≤ 140.

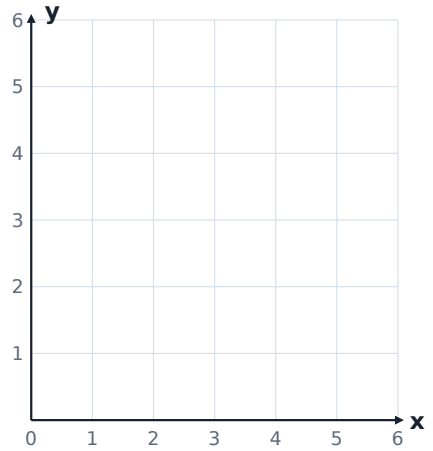
GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph the system $x + y \leq 5$, $x \geq 0$, and $y \geq 0$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S34-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices (0, 0), (5, 0), and (0, 5), shaded, with solid sides

GRADING NOTE Any correct graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is in the solution region of $y \geq 2x$ and $y \leq 6$?

A. (4, 1)

B. (1, 7)

C. (3, 5)

D. (1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 4)

GRADING NOTE $4 \geq 2$ and $4 \leq 6$. (3, 5) fails $5 \geq 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (0, 0) satisfies $x \geq 0$ and $y \geq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

1040

QUESTION ID
S34-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $0 \geq 0$ is true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q4/5

1040

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The point (2, 5) satisfies $x + y < 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 2 + 5 = 7, which is not less than 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.12 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-CED.A.3

BACKWARD 2

Solve a system of linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + y = 45$

$15x + 10y = 500$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S34-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(10, 35)

GRADING NOTE Any equivalent form. $15x + 450 - 10x = 500$, so $x = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$2x + y = 12$

$x + y = 8$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(4, 4)

GRADING NOTE Any equivalent form. Subtract: x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve.

$x + 3y = 60$

$x + y = 40$

A. (30, 10)

B. (15, 15)

C. (10, 30)

D. (20, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (30, 10)

GRADING NOTE Subtract: 2y = 20.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(10, 35) is a solution of x + y = 45.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 + 35 = 45.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q4/5

True or false?

The system x + y = 10 and x + y = 12 has one solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE No pair can have both sums: no solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.C.6 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-CED.A.3

FORWARD 1

Use constraints to optimize a design • High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A feasible region has vertices (0, 0), (60, 0), and (0, 20).

Find the greatest value of P = 2m + 7b on the region and where it occurs.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

1049

QUESTION ID
S34-F1-Q1

TYPE
Short answer

CORRECT ANSWER
140, at (0, 20)

GRADING NOTE Form required: the value and the point. P = 0, 120, and 140 at the vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q1/5

A rectangular pen uses at most 40 m of fence.

What dimensions give the greatest area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

1050

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 m by 10 m (area 100 m²)

GRADING NOTE Form required: the dimensions. A = w(20 − w) is greatest at w = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q2/5

The cost C = 3x + 2y is checked at the vertices (0, 6), (2, 2), and (6, 0) of a region.

Where is the cost least?

A. (2, 2)

B. (0, 0)

C. (6, 0)

D. (0, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

1051

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 2)

GRADING NOTE C = 12, 10, and 18 at the vertices. (0, 0) is not in the region.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q3/5

True or false?

For a linear expression on a polygonal region, the greatest value occurs at a vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

1052

QUESTION ID
S34-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the basis of linear programming.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q4/5

True or false?

Of all rectangles with area 36, a square has the smallest perimeter, so its side is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

1053

QUESTION ID
S34-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The side is $\sqrt{36} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-CED.A.3

FORWARD 2

Use expected values to compare decisions • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions • HSS-MD.B.7

ANSWER KEY Reduced copy of the question page, with its answer

Plan A earns $100 with probability 0.6 and $0 otherwise.
Plan B earns $70 for sure.

Which plan has the greater expected value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5

1055

QUESTION ID
S34-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Plan B: $70 compared with $60 for Plan A.

GRADING NOTE Form required: the plan and both expected values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Guessing on a 4-choice question earns 1 point if right and loses 0.25 point if wrong.

What is the expected score from guessing?

Expected score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5

1056

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.0625 point

GRADING NOTE Any equivalent form, such as $\frac{1}{16}$. $0.25(1) + 0.75(-0.25) = 0.0625$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q2/5

A $5 warranty pays $100 with probability 0.04.

What is the expected net value of buying it?

A. −$5

B. $1

C. $4

D. −$1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5

1057

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −$1

GRADING NOTE 100(0.04) − 5 = −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q3/5

True or false?

A choice with the higher expected value can still lose money on a single trial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5

1058

QUESTION ID
S34-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Expected value describes the long-run average.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q4/5

True or false?

If a game has an expected value of 0, a player cannot lose money playing it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5

1059

QUESTION ID
S34-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Individual plays can lose; only the long-run average is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5
High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Using Probability to Make Decisions
HSS-MD.B.7 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-IF.A.1 • INTERPRETING FUNCTIONS

Understand a function as a rule that assigns exactly one output to each input, with a domain and a range

MAIN	Algebra I main question	**HSF-IF.A.1**
BACKWARD 1	Decide whether a relation is a function	**8.F.A.1**
BACKWARD 2	Check whether points are on the graph of an equation	**HSA-REI.D.10**
FORWARD 1	Describe transformations of the plane as functions of points	**HSG-CO.A.2**
FORWARD 2	Restrict a domain to make a function invertible	**HSF-BF.B.4.d**

SET 35 • HSF-IF.A.1

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Does the relation {(−2, 4), (0, 1), (2, 4), (3, −1)} define y as a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

1062

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each input has exactly one output. (Two inputs may share the output 4.)

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the domain and the range of $f(x) = x^2 + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

1063

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

Domain: all real numbers. Range: $y \geq 1$.

GRADING NOTE Form required: both the domain and the range. $x^2 \geq 0$, so $x^2 + 1 \geq 1$, and every value 1 or more is reached.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation does NOT define y as a function of x?

A. $y = x^2$

B. $x = y^2$

C. $y = 2^x$

D. $y = |x|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

1064

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $x = y^2$

GRADING NOTE For $x = y^2$, the input $x = 4$ has two outputs, $y = 2$ and $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

f(x) = √(x − 3). Over the real numbers, what is the domain of f?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

1065

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

x ≥ 3

GRADING NOTE Form required: an inequality (or interval [3, ∞)). The expression under the square root must not be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If f is a function and f(2) = 5, then f(2) can also equal 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

1066

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A function assigns exactly one output to each input.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 1

Decide whether a relation is a function • Grade 8 • Functions • 8.F.A.1

A table pairs x = 1, 2, 3, 2 with y = 4, 5, 6, 7. Is y a function of x? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

No. The input 2 has two outputs, 5 and 7.

GRADING NOTE Form required: yes or no with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Each button on a vending machine gives exactly one kind of snack. Is the snack a function of the button? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

1069

GRADING NOTE Form required: yes or no with a reason.

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. Each input (button) gives exactly one output (snack).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q2/5

Which set of ordered pairs is a function?

A. {(5, 1), (5, 2), (6, 3)}

B. {(0, 0), (0, 1)}

C. {(1, 3), (2, 3), (3, 3)}

D. {(1, 2), (1, 3), (2, 4)}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. {(1, 3), (2, 3), (3, 3)}

GRADING NOTE Every input in the first set appears once. Each other set repeats an input with a different output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q3/5

True or false?

Two different inputs of a function can have the same output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE For example, $f(x) = x^2$ gives $f(2) = f(-2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A function can pair the input 4 with two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Each input has exactly one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

BACKWARD 2

Check whether points are on the graph of an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

ANSWER KEY Reduced copy of the question page, with its answer

Is (−2, 5) on the graph of $y = x^2 + 1$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. $(-2)^2 + 1 = 5$.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (a, 10) is on the graph of y = 2x + 4.

Find a.

a = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 2a + 4 = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which point is on the graph of $y = \sqrt{x}$?

A. (9, −3)

B. (−9, 3)

C. (9, 3)

D. (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (9, 3)

GRADING NOTE $\sqrt{9} = 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of y = 4 is the set of all points with y-coordinate 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

1077

QUESTION ID
S35-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every such point makes the equation true.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

(1, 3) is on the graph of $y = x^2 + 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $1^2 + 3 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 1

Describe transformations of the plane as functions of points • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2

ANSWER KEY Reduced copy of the question page, with its answer

The translation T maps (x, y) to (x + 3, y − 2).

Find T(4, 1).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

1080

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, −1)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A reflection across the y-axis maps (x, y) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

1081

QUESTION ID
S35-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(−x, y)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which transformation does NOT preserve distance?

A. $(x, y) \to (x + 5, y)$

B. $(x, y) \to (-x, y)$

C. $(x, y) \to (y, x)$

D. $(x, y) \to (2x, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

1082

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (2x, y)

GRADING NOTE (x, y) → (2x, y) stretches horizontal distances. The others are a translation and two reflections.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q3/5

True or false?

A transformation of the plane is a function: each input point has exactly one image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

1083

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE That is what makes it a function.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q4/5

True or false?

A rotation of 90° counterclockwise about the origin maps (1, 0) to (0, −1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

1084

QUESTION ID
S35-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It maps (1, 0) to (0, 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S35-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-IF.A.1

FORWARD 2

Restrict a domain to make a function invertible • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d

Explain why $f(x) = x^2$, with all real numbers as its domain, has no inverse function.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

1086

QUESTION ID
S35-F2-Q1

TYPE
Short answer

CORRECT ANSWER
f(2) = f(−2) = 4, so an inverse would have to send 4 to two different outputs.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Restrict the domain of $f(x) = (x - 1)^2$ so that f has an inverse, and write the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

1087

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$x \geq 1$; $f^{-1}(x) = 1 + \sqrt{x}$

GRADING NOTE Form required: the restriction and the inverse. The restriction $x \leq 1$ with $f^{-1}(x) = 1 - \sqrt{x}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q2/5

Which function is one-to-one on all real numbers?

A. $f(x) = x^2 - 4$

B. $f(x) = x^3$

C. $f(x) = |x|$

D. $f(x) = x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

1088

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3$

GRADING NOTE x^3 takes each output once. The others give the same output for x and −x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S35-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^2$ with domain $x \geq 0$ has the inverse $f^{-1}(x) = \sqrt{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

1089

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE On $x \geq 0$, $\sqrt{(x^2)} = x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Every function has an inverse function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

1090

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Only one-to-one functions do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.d • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-IF.A.2 • INTERPRETING FUNCTIONS

Use function notation, evaluate functions, and interpret statements in function notation in context

MAIN	Algebra I main question	**HSF-IF.A.2**
BACKWARD 1	Evaluate expressions for given values	**6.EE.A.2.c**
BACKWARD 2	Solve linear equations	**HSA-REI.B.3**
FORWARD 1	Evaluate compositions of functions	**HSF-BF.A.1.c**
FORWARD 2	Read values of an inverse function	**HSF-BF.B.4.c**

SET 36 • HSF-IF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x^2 - 3x + 1.$

Find $f(-2)$.

$f(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

1093

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 2(4) + 6 + 1 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet t seconds after it is hit.

Find h(1), and explain what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

1094

QUESTION ID
S36-M-Q2

TYPE
Short answer

CORRECT ANSWER
h(1) = 48: one second after it is hit, the ball is 48 feet high.

GRADING NOTE Form required: the value and its meaning (with units).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 5x - 7$.

Find the value of x for which $g(x) = 18$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

1095

QUESTION ID

S36-M-Q3

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $5x - 7 = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

C(n) is the cost in dollars of n tickets.

What does C(4) = 60 mean?

A. 4 tickets cost $60.

B. C equals 4 times 60.

C. Each ticket costs $60.

D. 60 tickets cost $4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

1096

QUESTION ID

S36-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4 tickets cost $60.

GRADING NOTE The input is the number of tickets and the output is the cost.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q4/5

True or false?

If $f(x) = x^2$, then $f(a + 1) = a^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

1097

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(a + 1) = (a + 1)^2 = a^2 + 2a + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 1

Evaluate expressions for given values • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $2x^2 - 3x + 1$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S36-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. $18 - 9 + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $-16t^2 + 64t$ when $t = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

64

GRADING NOTE Any equivalent form. $-64 + 128$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q2/5

Evaluate $3(a - 4)^2$ when $a = 6$.

A. 144

B. 12

C. 6

D. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

GRADING NOTE $3(2)^2 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

When $x = -1$, $x^2 - x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

1102

GRADING NOTE 1 + 1 = 2.

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When $n = 5$, $2n^2 = 100$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $2(25) = 50$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

BACKWARD 2

Solve linear equations • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

ANSWER KEY Reduced copy of the question page, with its answer

Solve 5x − 7 = 18.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 3(x + 2) = 2x − 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−10

GRADING NOTE Any equivalent form. 3x + 6 = 2x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve 0.5x + 3 = 11.

A. x = 7

B. x = 28

C. x = 16

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 16

GRADING NOTE 0.5x = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = -2$ is the solution of $4x + 9 = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

1108

QUESTION ID
S36-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $-8 + 9 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The solution of $7 - x = 10$ is $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $-x = 3$, so $x = -3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 1

Evaluate compositions of functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x + 4$ and $g(x) = 3x$.

Find $f(g(2))$.

$f(g(2)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

1111

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10

GRADING NOTE Any equivalent form. g(2) = 6 and f(6) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ and $g(x) = x - 3$.

Write $f(g(x))$.

$f(g(x)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

1112

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 3)^2$

GRADING NOTE Any equivalent form, such as $x^2 - 6x + 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x$ and $g(x) = x + 1$.

Find $g(f(5))$.

A. 11

B. 10

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

1113

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 11

GRADING NOTE $f(5) = 10$ and $g(10) = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q3/5

True or false?

If f(x) = x − 1 and g(x) = x + 1, then f(g(x)) = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

1114

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE (x + 1) − 1 = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q4/5

True or false?

If $f(x) = 2x$ and $g(x) = x^2$, then $f(g(3)) = 36$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

1115

QUESTION ID
S36-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $g(3) = 9$ and $f(9) = 18$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-IF.A.2

FORWARD 2

Read values of an inverse function • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

A function f has f(1) = 3, f(2) = 5, f(3) = 7, and f(4) = 9.

Find $f^{-1}(7)$.

$f^{-1}(7)$ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

1117

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. f(3) = 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x + 1$ is invertible.

Find $f^{-1}(11)$.

$f^{-1}(11) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

1118

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

GRADING NOTE Any equivalent form. $2x + 1 = 11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q2/5

If f(4) = 10, which statement is true?

A. $f^{-1}(10) = 4$

B. $f^{-1}(4) = 10$

C. $f(10) = 4$

D. $f^{-1}(10) = \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

1119

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $f^{-1}(10) = 4$

GRADING NOTE The inverse reverses the pair (4, 10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q3/5

True or false?

The graph of f^{-1} is the reflection of the graph of f across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

1120

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each point (a, b) becomes (b, a).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q4/5

True or false?

$f^{-1}(x)$ means $1 \div f(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

1121

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f^{-1} is the inverse function, not the reciprocal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S36-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 36 • FORWARD 2 • Q5/5

Recognize that sequences are functions whose domain is a set of integers

MAIN	Algebra I main question	**HSF-IF.A.3**
BACKWARD 1	Evaluate functions in function notation	**HSF-IF.A.2**
BACKWARD 2	Generate numerical patterns from rules	**5.OA.B.3**
FORWARD 1	Repeat a dilation and describe the lengths	**HSG-SRT.A.1**
FORWARD 2	Find the sum of a finite geometric series	**HSA-SSE.B.4**

SET 37 • HSF-IF.A.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(1) = 3$ and $f(n) = f(n - 1) + 4$ for $n \geq 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

1124

QUESTION ID

S37-M-Q1

TYPE

Short answer

CORRECT ANSWER

19

GRADING NOTE Any equivalent form. 3, 7, 11, 15, 19.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence is defined by $f(0) = f(1) = 1$ and $f(n + 1) = f(n) + f(n - 1)$ for $n \geq 1$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

1125

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

8

GRADING NOTE Any equivalent form. 1, 1, 2, 3, 5, 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula for the sequence 5, 10, 20, 40, ..., with f(1) = 5.

f(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

1126

QUESTION ID

S37-M-Q3

TYPE

Short answer

CORRECT ANSWER

$5 \cdot 2^{n-1}$

GRADING NOTE Any equivalent form, such as $2.5 \cdot 2^{n}$. Each term is 2 times the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A sequence f(n) starts with f(1).

What is its domain?

A. The positive integers 1, 2, 3, ...

B. The numbers from 0 to 1

C. All real numbers

D. All integers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

1127

QUESTION ID

S37-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The positive integers 1, 2, 3, ...

GRADING NOTE A sequence is defined only at the term numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q4/5

True or false?

The graph of the sequence f(n) = 2n + 1 for n = 1, 2, 3, ... is a solid line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

1128

QUESTION ID

S37-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The domain is the positive integers, so the graph is separate points: (1, 3), (2, 5), (3, 7), ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 37 • MAIN • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 1

Evaluate functions in function notation • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 3n - 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

11

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2^n$.

Find f(5).

f(5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = n^2 + 1$.

Find f(3).

A. 7

B. 10

C. 16

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

GRADING NOTE $9 + 1 = 10$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q3/5

True or false?

If f(n) = 10 − n, then f(10) = 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 10 − 10 = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q4/5

True or false?

If f(n) = 4n, then f(3) = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 \cdot 3 = 12$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

BACKWARD 2

Generate numerical patterns from rules • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Start at 3 and add 4 each time.

Write the first 5 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3, 7, 11, 15, 19

GRADING NOTE Form required: the five terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Start at 1 and multiply by 3 each time.

What is the 4th term?

4th term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

27

GRADING NOTE Any equivalent form. 1, 3, 9, 27.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6.

How do their third terms compare?

A. A's is 3 times B's.

B. B's is 4 more than A's.

C. B's is 3 times A's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. B's is 3 times A's.

GRADING NOTE A: 0, 2, 4. B: 0, 6, 12. 12 = 3 × 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The rule "start at 2 and add 5" gives 2, 7, 12, 17.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each term is 5 more than the one before.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q4/5

True or false?

The rule "start at 3 and double" gives 3, 5, 7, 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Doubling gives 3, 6, 12, 24.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 1

Repeat a dilation and describe the lengths • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A segment of length 3 is dilated by scale factor 2, then the image is dilated by 2 again, and so on.

Write the length after n dilations.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5

1142

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$3 \cdot 2^n$

GRADING NOTE Any equivalent form. Each dilation multiplies the length by 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q1/5

A dilation with center (0, 0) and scale factor 3 maps (2, −1) to which point?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5

1143

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −3)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q2/5

A dilation with scale factor $\frac{1}{2}$ maps a segment of length 10 to a segment of what length?

A. 20

B. 9.5

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5

1144

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

GRADING NOTE $10 \cdot \frac{1}{2} = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5

1145

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is a property of dilations.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 doubles the area of a figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5

1146

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Areas are multiplied by $2^2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-IF.A.3

FORWARD 2

Find the sum of a finite geometric series • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 5 terms of 2, 6, 18, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

1148

QUESTION ID

S37-F2-Q1

TYPE

Short answer

CORRECT ANSWER

242

GRADING NOTE Any equivalent form. $2(3^5 - 1) \div (3 - 1) = 242$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a formula for the sum of the first n terms of 5, 10, 20, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

1149

QUESTION ID
S37-F2-Q2

TYPE
Short answer

CORRECT ANSWER
$5(2^n - 1)$

GRADING NOTE Any equivalent form. a = 5 and r = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 1 + 4 + 16 + 64?

A. 256

B. 85

C. 84

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

1150

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 85

GRADING NOTE $(4^4 - 1) \div 3 = 85$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q3/5

True or false?

The sum of the first 4 terms of 1, 2, 4, 8, ... is 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

1151

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1 + 2 + 4 + 8 = 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q4/5

True or false?

A geometric series with first term 4, ratio 1, and 5 terms has sum 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

1152

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 + 4 + 4 + 4 + 4 = 20$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of graphs and tables of functions in context

MAIN	Algebra I main question	**HSF-IF.B.4**
BACKWARD 1	Describe a function qualitatively from a situation	**8.F.B.5**
BACKWARD 2	Use vertex form to find a maximum or minimum	**HSA-SSE.B.3.b**
FORWARD 1	Identify key features of polynomial functions	**HSF-IF.C.7.c**
FORWARD 2	Identify amplitude, midline, and period of a periodic model	**HSF-TF.B.5**

SET 38 • HSF-IF.B.4

MAIN

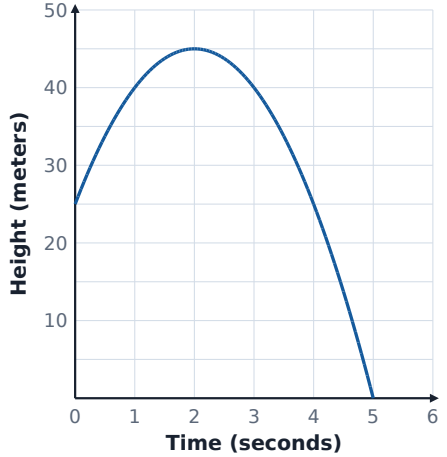
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows a ball's height h(t) in meters, t seconds after it is thrown from a balcony.

What is the maximum height, and when does it occur?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

1155

QUESTION ID
S38-M-Q1

TYPE
Short answer

CORRECT ANSWER
45 meters, at t = 2 seconds

GRADING NOTE Form required: the height and the time, with units.

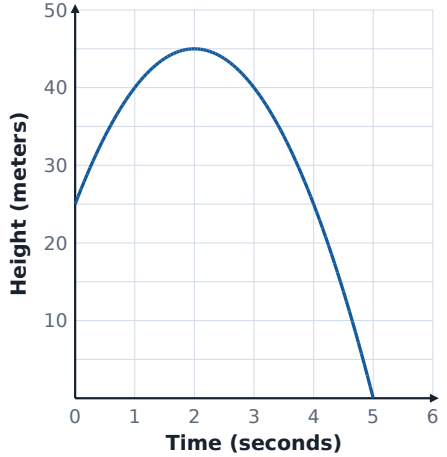
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the h-intercept of the graph, and what does it mean?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

1156

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

25: the ball was thrown from 25 meters above the ground.

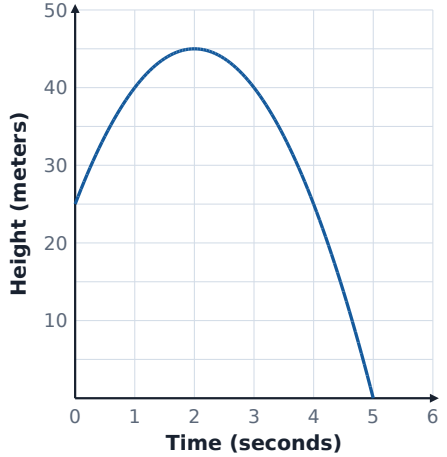
GRADING NOTE Form required: the value and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q2/5

When does the ball reach the ground? Over what interval is its height increasing?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

1157

QUESTION ID

S38-M-Q3

TYPE

Short answer

CORRECT ANSWER

At t = 5 seconds; the height increases for 0 < t < 2.

GRADING NOTE Form required: both parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q3/5

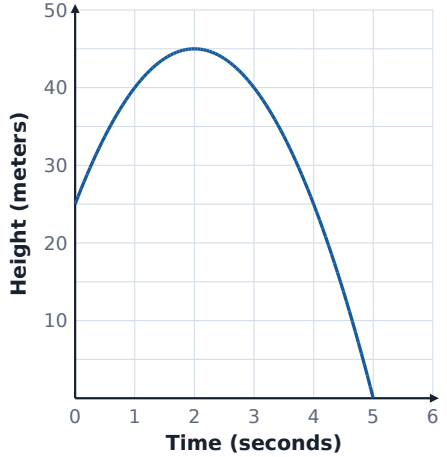
Over what interval is the ball's height decreasing?

A. 0 < t < 2

B. 0 < t < 5

C. 2 < t < 5

D. 25 < t < 45



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

1158

QUESTION ID

S38-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2 < t < 5

GRADING NOTE After the highest point at t = 2, the height falls until the ball lands at t = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q4/5

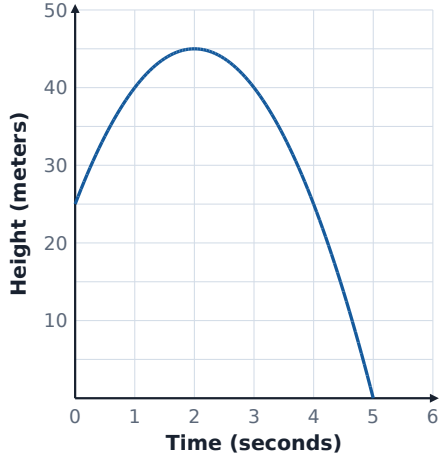
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The ball is 25 meters high twice: at t = 0 and at t = 4.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

1159

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph is symmetric about t = 2, and 0 and 4 are each 2 seconds from it.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 38 • MAIN • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 1

Describe a function qualitatively from a situation • Grade 8 • Functions • 8.F.B.5

A car speeds up, drives at a constant speed, then slows to a stop.

Describe the graph of its speed over time.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

It rises, then stays level, then falls to 0.

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q1

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of water level is horizontal from t = 2 to t = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The water level did not change from t = 2 to t = 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q2/5

Which situation has a graph that first increases and then decreases?

A. Savings with a steady deposit each week

B. The height of a ball thrown up that comes back down

C. The distance a parked car travels

D. The height of a burning candle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The height of a ball thrown up that comes back down

GRADING NOTE The ball rises and then falls. The others only decrease, stay the same, or only increase.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q3/5

True or false?

On a distance-time graph, a steeper segment means a greater speed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE More distance in the same time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q4

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q4/5

True or false?

A decreasing graph always means the quantity is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A quantity can decrease while staying positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q5

Grade 8 • Functions • 8.F.B.5 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

BACKWARD 2

Use vertex form to find a maximum or minimum • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Write $-5t^2 + 20t + 25$ in the form $a(t - h)^2 + k$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$-5(t - 2)^2 + 45$

GRADING NOTE Form required: $a(t - h)^2 + k$. $-5(t^2 - 4t) + 25 = -5(t - 2)^2 + 20 + 25$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q1/5

What is the maximum value of $-(x - 3)^2 + 7$, and for what x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7, at x = 3

GRADING NOTE Form required: the value and the x-value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q2/5

What is the minimum value of $2(x + 1)^2 - 8$?

A. -8

B. -1

C. 2

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −8

GRADING NOTE $2(x + 1)^2 \geq 0$, and it is 0 at $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$-5(t-2)^2 + 45$ is greatest when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

1170

QUESTION ID
S38-B2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE The squared term is 0 there.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

The minimum value of $(x - 4)^2 + 1$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum value is 1, at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 1

Identify key features of polynomial functions • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

Describe the end behavior of $f(x) = -2x^3 + x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

1173

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

As $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$.

GRADING NOTE Any equivalent description. The leading term $-2x^3$ decides the end behavior.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q1/5

Find the zeros of f(x) = x(x − 2)(x + 3).

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

1174

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 2, and −3

GRADING NOTE Form required: all three zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q2/5

Which function rises on the right and falls on the left?

A. $f(x) = -x^4$

B. $f(x) = x^3 - 4x$

C. $f(x) = x^2$

D. $f(x) = -x^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

1175

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $f(x) = x^3 - 4x$

GRADING NOTE An odd degree with a positive leading coefficient.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q3/5

True or false?

A polynomial function of degree 4 has at most 4 real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

1176

QUESTION ID
S38-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE A degree-n polynomial has at most n zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q4/5

True or false?

The graph of $f(x) = x^4$ falls on both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

1177

QUESTION ID

S38-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x^4 is never negative; the graph rises on both ends.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-IF.B.4

FORWARD 2

Identify amplitude, midline, and period of a periodic model • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel seat's height is $h(t) = 30 - 25\cos(\pi t \div 5)$ meters after t minutes.

What are the greatest and least heights?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

1179

QUESTION ID
S38-F2-Q1

TYPE
Short answer

CORRECT ANSWER
55 meters and 5 meters

GRADING NOTE Form required: both heights. cos is between −1 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the period of h(t) = 30 − 25cos(πt ÷ 5)?

Period: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

1180

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 minutes

GRADING NOTE Any equivalent form. 2π ÷ (π ÷ 5) = 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S38-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q2/5

What is the midline of y = 4sin x + 2?

A. y = 4

B. y = 2

C. y = 6

D. y = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

1181

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. y = 2

GRADING NOTE The graph goes from −2 to 6, centered on y = 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q3/5

True or false?

The amplitude of y = 3sin x is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

1182

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The graph goes 3 units above and below its midline.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q4/5

True or false?

The function y = sin x has period π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

1183

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The period is 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-IF.B.5 • INTERPRETING FUNCTIONS

Relate the domain of a function to the quantity it describes

MAIN	Algebra I main question	**HSF-IF.B.5**
BACKWARD 1	Decide whether a solution is viable	**HSA-CED.A.3**
BACKWARD 2	Find where a quadratic model equals 0	**HSA-REI.B.4.b**
FORWARD 1	Use trigonometric ratios as functions of acute angles	**HSG-SRT.C.6**
FORWARD 2	Extend trigonometric functions to all real numbers with the unit ci…	**HSF-TF.A.2**

SET 39 • HSF-IF.B.5

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

n(k) = 12k gives the number of eggs in k full cartons.
The store has 30 cartons.

What is a reasonable domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

1186

QUESTION ID
S39-M-Q1

TYPE
Short answer

CORRECT ANSWER
The whole numbers 0, 1, 2, ..., 30

GRADING NOTE Form required: the domain, stated as whole numbers from 0 to 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$h(t) = -16t^2 + 64t$ gives a ball's height in feet from the moment it is hit until it lands.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

1187

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

$0 \le t \le 4$

GRADING NOTE Form required: an inequality (or interval). $h(t) = -16t(t - 4) = 0$ at $t = 0$ and $t = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q2/5

A taxi fare is C(m) = 3 + 2.5m dollars for m miles, with a limit of 40 miles per ride.

What is the domain?

A. All real numbers

B. m ≥ 3

C. 3 ≤ m ≤ 103

D. 0 ≤ m ≤ 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

1188

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0 ≤ m ≤ 40

GRADING NOTE The input is miles, from 0 to the 40-mile limit. 3 ≤ C ≤ 103 is the range.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

A rectangle has perimeter 20. Its area is $A(w) = w(10 - w)$, where w is the width.

What is the domain?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

1189

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

$0 < w < 10$

GRADING NOTE Form required: an inequality. Both w and the length $10 - w$ must be positive.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q4/5

True or false?

P(n) = 50n gives the number of seats on n buses.
n = 2.5 is in a reasonable domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

1190

QUESTION ID
S39-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The number of buses must be a whole number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 1

Decide whether a solution is viable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

Solving a problem gives 7.4 buses needed to carry every student.

What is the viable answer? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 buses: the number must be a whole number, and 7 buses are not enough.

GRADING NOTE Form required: the number and a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The number of tickets t must satisfy $5t \leq 60$ and $t \geq 0$.

What whole numbers of tickets are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0, 1, 2, ..., 12

GRADING NOTE Any equivalent form, such as the whole numbers from 0 to 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value is NOT viable for the number of people in a room?

A. 40

B. 0

C. 12

D. −3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −3

GRADING NOTE A count cannot be negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A length of 0.25 meter can be a viable solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Lengths can be fractions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

w(10 − w) = 24 has solutions w = 4 and w = 6, but only one of them is a viable width.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both are viable: width 4 with length 6, or width 6 with length 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

BACKWARD 2

Find where a quadratic model equals 0 • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 or t = 4

GRADING NOTE Form required: both solutions. $-16t(t - 4) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $w(10 - w) = 0$.

Solutions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$w = 0$ or $w = 10$

GRADING NOTE Form required: both solutions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $-16t^2 + 64t + 80 = 0$.

A. $t = 4$ or $t = -2$

B. $t = 5$ or $t = -1$

C. $t = 5$ only

D. $t = -5$ or $t = 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $t = 5$ or $t = -1$

GRADING NOTE $t^2 - 4t - 5 = (t - 5)(t + 1) = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$t = 4$ is a solution of $t^2 - 4t = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $16 - 16 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = -9$ has no real solution.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 1

Use trigonometric ratios as functions of acute angles • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

ANSWER KEY Reduced copy of the question page, with its answer

Right triangle ABC has its right angle at C, with AC = 4, BC = 3, and AB = 5.

Find sin A.

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5

1204

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE Any equivalent form, such as 0.6. sin A = opposite ÷ hypotenuse = BC ÷ AB.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q1**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

For which angle measures θ is sin θ defined as a ratio of sides of a right triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5

1205

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$0° < \theta < 90°$

GRADING NOTE Form required: an inequality. Only the acute angles of a right triangle are used.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q2/5

In a right triangle, $\cos A = \frac{12}{13}$.

What is sin A?

A. $\frac{5}{13}$

B. $\frac{12}{5}$

C. $\frac{1}{13}$

D. $\frac{13}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5

1206

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5/13

GRADING NOTE The sides are 5, 12, 13, so the opposite side is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q3**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q3/5

True or false?

In a right triangle, sin A = cos B when A and B are the two acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5

1207

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The side opposite A is adjacent to B.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q4/5

True or false?

For an acute angle θ, sin θ can be greater than 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5

1208

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The opposite side is shorter than the hypotenuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-IF.B.5

FORWARD 2

Extend trigonometric functions to all real numbers with the unit circle • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Find cos π.

cos π = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

1210

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point at angle π is (−1, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find sin(3π ÷ 2).

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

1211

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. The point is (0, −1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q2/5

With the unit circle, what is the domain of f(θ) = sin θ?

A. All real numbers

B. $0 < \theta < \pi \div 2$

C. $\theta \geq 0$

D. $0 \leq \theta \leq 2\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

1212

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. All real numbers

GRADING NOTE Any real number of radians, positive or negative, gives a point on the unit circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q3/5

True or false?

A negative radian measure gives a clockwise rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

1213

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Positive angles turn counterclockwise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q4/5

True or false?

sin θ is defined only for acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

1214

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The unit circle defines it for every real number.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-IF.B.6 • INTERPRETING FUNCTIONS

Calculate, interpret, and estimate the average rate of change of a function over an interval

MAIN	Algebra I main question	**HSF-IF.B.6**
BACKWARD 1	Find the rate of change of a linear function	**8.F.B.4**
BACKWARD 2	Compute unit rates with fractions	**7.RP.A.1**
FORWARD 1	Find average rates of change of periodic models	**HSF-TF.B.5**
FORWARD 2	Analyze rational functions: rates of change and asymptotes	**HSF-IF.C.7.d**

SET 40 • HSF-IF.B.6

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2 + 2x$.

Find the average rate of change from $x = 1$ to $x = 4$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

1217

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $(f(4) - f(1)) \div (4 - 1) = (24 - 3) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town had 12,000 people in 2010 and 15,600 in 2022.

Find the average rate of change, with units.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

1218

QUESTION ID
S40-M-Q2

TYPE
Short answer

CORRECT ANSWER
300 people per year

GRADING NOTE Form required: the rate with its units. 3,600 ÷ 12 = 300.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The table gives the distance d a cart has traveled after t seconds.

Over which 2-second interval was its average speed greatest? Give that speed.

t (seconds)	0	2	4	6
d (meters)	0	30	100	150

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

1219

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

From t = 2 to t = 4, at 35 meters per second

GRADING NOTE Form required: the interval and the speed. The speeds are 15, 35, and 25 m/s.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the average rate of change of $g(x) = 3^x$ from $x = 0$ to $x = 2$?

A. 4

B. 3

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

1220

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4

GRADING NOTE $(9 - 1) \div 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q4/5

True or false?

A linear function has the same average rate of change over every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

1221

QUESTION ID
S40-M-Q5

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Its average rate of change is always its slope.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 40 • MAIN • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 1

Find the rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the rate of change of the line through (2, 5) and (6, 17).

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 12 ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q1

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q1/5

A tank had 50 liters at t = 0 and 20 liters at t = 10 minutes, draining steadily.

What is the rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

−3 liters per minute

GRADING NOTE Any equivalent form, with units. (20 − 50) ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q2/5

What is the rate of change of $y = 7 - 4x$?

A. 7

B. -4

C. $-\frac{4}{7}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −4

GRADING NOTE The coefficient of x is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q3

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A negative rate of change means the quantity is decreasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Outputs fall as inputs rise.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q4

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q4/5

True or false?

The rate of change from (0, 3) to (2, 3) is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(3 - 3) \div 2 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q5

Grade 8 • Functions • 8.F.B.4 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

BACKWARD 2

Compute unit rates with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A runner goes $\frac{3}{4}$ mile in $\frac{1}{8}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 miles per hour

GRADING NOTE Any equivalent form. $\frac{3}{4} \div \frac{1}{8} = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A car uses 12 gallons to go 300 miles.

How many miles per gallon?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A recipe uses $\frac{2}{3}$ cup of sugar for $\frac{1}{3}$ of a batch.

How much sugar is used per batch?

A. $\frac{1}{2}$ cup

B. 1 cup

C. 2 cups

D. $\frac{2}{9}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2 cups

GRADING NOTE $\frac{2}{3} \div \frac{1}{3} = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

120 km in 1.5 hours is 80 km per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 120 ÷ 1.5 = 80.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$18 for 4 pounds is $4 per pound.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 18 ÷ 4 = $4.50 per pound.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 1

Find average rates of change of periodic models • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The temperature is $T(h) = 20 + 8\sin(\pi h \div 12)$ °C, h hours after 6 a.m.

Find $T(0)$ and $T(6)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

1235

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$T(0) = 20$ °C and $T(6) = 28$ °C

GRADING NOTE Form required: both values. $\sin 0 = 0$ and $\sin(\pi \div 2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Using T(0) = 20 and T(6) = 28, find the average rate of change from h = 0 to h = 6.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

1236

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{4}{3}$ °C per hour (about 1.33)

GRADING NOTE Any equivalent form, with units. 8 ÷ 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q2/5

What is the average rate of change of $y = \sin x$ from $x = 0$ to $x = 2\pi$?

A. 1

B. -1

C. 2π

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

1237

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 0

GRADING NOTE $\sin 2\pi = \sin 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q3/5

True or false?

For a periodic function with period 12, the average rate of change over any interval of length 12 is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

1238

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE f(a + 12) = f(a).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q4/5

True or false?

The average rate of change of y = sin x from 0 to π ÷ 2 is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

1239

QUESTION ID
S40-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 1 ÷ (π ÷ 2) = 2 ÷ π ≈ 0.64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-IF.B.6

FORWARD 2

Analyze rational functions: rates of change and asymptotes • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 12 \div x$.

Find the average rate of change from $x = 2$ to $x = 6$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

1241

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

−1

GRADING NOTE Any equivalent form. $(2 - 6) \div (6 - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Give the vertical and horizontal asymptotes of $f(x) = 1 \div (x - 3) + 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

1242

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Vertical x = 3; horizontal y = 2

GRADING NOTE Form required: both asymptotes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q2/5

Where is f(x) = (x + 1) ÷ (x − 4) undefined?

A. x = −4

B. x = 1

C. x = −1

D. x = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

1243

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 4

GRADING NOTE The denominator is 0 at x = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q3**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q3/5

True or false?

The graph of $y = 1 \div x$ gets closer to the x-axis as x gets large.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

1244

QUESTION ID
S40-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 1 ÷ x approaches 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 1 \div x$ crosses the y-axis at (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

1245

QUESTION ID
S40-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $1 \div 0$ is undefined, so the graph never meets the y-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph linear functions and show their intercepts

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Find slope and write y = mx + b	**8.EE.B.6**
BACKWARD 2	Write linear equations in two variables from a situation	**HSA-CED.A.2**
FORWARD 1	Find a point that partitions a segment in a given ratio	**HSG-GPE.B.6**
FORWARD 2	Find and verify inverses of linear functions	**HSF-BF.B.4.b**

SET 41 • HSF-IF.C.7.a

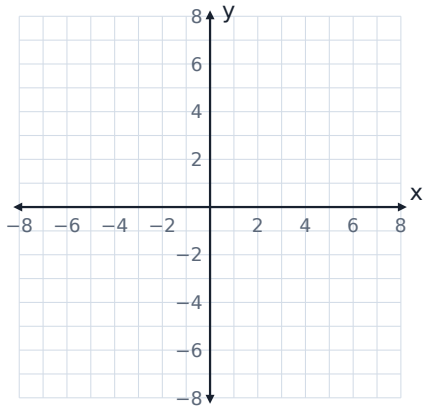
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = -\frac{2}{3}x + 4$ on the coordinate plane.

Give the x-intercept and the y-intercept.



Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

1248

QUESTION ID

S41-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A straight line through (0, 4), (3, 2), and (6, 0)

Written: x-intercept (6, 0); y-intercept (0, 4)

GRADING NOTE Form required: both intercepts. Both the graph and the intercepts are graded. $-\frac{2}{3}x + 4 = 0$ gives $x = 6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x- and y-intercepts of 3x − 5y = 15.

Intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

1249

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 0) and (0, −3)

GRADING NOTE Form required: both intercepts. Set y = 0, then x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Write the equation of the line through (−2, 7) and (4, −5) in slope-intercept form.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

1250

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

y = −2x + 3

GRADING NOTE Form required: slope-intercept form, y = mx + b. The slope is −12 ÷ 6 = −2, and 7 = −2(−2) + b.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q3/5

Which describes the graph of y = −3?

A. A line with slope 1 through (−3, −3)

B. A vertical line through (−3, 0)

C. A line with slope −3 through the origin

D. A horizontal line through (0, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

1251

QUESTION ID

S41-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. A horizontal line through (0, −3)

GRADING NOTE y is −3 for every x, and the slope is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $2x + y = 6$ has x-intercept 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

1252

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At y = 0, 2x = 6, so the x-intercept is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 41 • MAIN • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 1

Find slope and write y = mx + b • Grade 8 • Expressions & Equations • 8.EE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the slope of the line through (1, 2) and (3, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. 6 ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q1/5

A line has slope 2 and y-intercept −1.

Write its equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x − 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 5x - 4$?

A. $\frac{4}{5}$

B. 4

C. 5

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. −4

GRADING NOTE $y = mx + b$ with $b = -4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Any two points on a line give the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Similar slope triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q4/5

True or false?

The line y = 4x passes through (1, 5).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 4(1) = 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

BACKWARD 2

Write linear equations in two variables from a situation • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A gym charges $25 to join plus $15 per month.

Write an equation for the cost C after m months.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 25 + 15m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A candle 20 cm tall burns 2 cm per hour.

Write an equation for its height h after t hours.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

h = 20 − 2t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Ana earns $8 per hour plus $5 in tips.

Which equation gives her earnings E for h hours?

A. E = 8h + 5

B. E = 5h + 8

C. E = 13h

D. E = 8(h + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. E = 8h + 5

GRADING NOTE 8 per hour, plus 5 once.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q3/5

True or false?

d = 60t gives the distance traveled at 60 miles per hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Distance = rate × time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q4/5

True or false?

y = 100 − 4x describes a quantity that grows by 4 each step.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 4 each step.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 1

Find a point that partitions a segment in a given ratio • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Find the midpoint of the segment from (−2, 3) to (6, 7).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

1266

QUESTION ID
S41-F1-Q1

TYPE
Short answer

CORRECT ANSWER
(2, 5)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q1/5

Find the point $\frac{1}{4}$ of the way from A(0, 0) to B(8, 12).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

1267

QUESTION ID
S41-F1-Q2

TYPE
Short answer

CORRECT ANSWER
(2, 3)

GRADING NOTE Any equivalent form. $\frac{1}{4}$ of (8, 12).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q2/5

Point P divides the segment from A(1, 2) to B(7, 11) in the ratio 1 : 2 (AP : PB).

Find P.

A. (5, 8)

B. (3, 5)

C. (4, 6.5)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

1268

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 5)

GRADING NOTE P is $\frac{1}{3}$ of the way: (1 + 2, 2 + 3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q3/5

True or false?

The midpoint of (0, 0) and (10, 4) is (5, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

1269

QUESTION ID
S41-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Average the coordinates.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q4**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q4/5

True or false?

The point $\frac{1}{3}$ of the way from (0, 0) to (9, 6) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

1270

QUESTION ID
S41-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It is (3, 2); (6, 4) is $\frac{2}{3}$ of the way.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F1-Q5**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.B.6 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-IF.C.7.a

FORWARD 2

Find and verify inverses of linear functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

Find the inverse of f(x) = 3x − 6.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

1272

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(x + 6) ÷ 3

GRADING NOTE Any equivalent form, such as $\frac{1}{3}x + 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Show that f(x) = 2x + 1 and g(x) = (x − 1) ÷ 2 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

1273

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = 2 · (x − 1) ÷ 2 + 1 = x, and g(f(x)) = (2x + 1 − 1) ÷ 2 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is the inverse of $f(x) = x \div 4 + 1$?

A. $f^{-1}(x) = 4x + 1$

B. $f^{-1}(x) = 4(x + 1)$

C. $f^{-1}(x) = 4x - 4$

D. $f^{-1}(x) = x \div 4 - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

1274

QUESTION ID
S41-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $f^{-1}(x) = 4x - 4$

GRADING NOTE Solve $x = y \div 4 + 1$ for y: $y = 4(x - 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q3/5

True or false?

If f(g(x)) = x and g(f(x)) = x for every x, then f and g are inverses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

1275

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = x + 5$ is $f^{-1}(x) = x + 5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

1276

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The inverse is $x - 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a • INTERPRETING FUNCTIONS

Graph quadratic functions and show intercepts, maxima, and minima

MAIN	Algebra I main question	**HSF-IF.C.7.a**
BACKWARD 1	Factor to find the zeros of a quadratic	**HSA-SSE.B.3.a**
BACKWARD 2	Interpret key features of a function	**HSF-IF.B.4**
FORWARD 1	Relate a parabola's equation to its focus and directrix	**HSG-GPE.A.2**
FORWARD 2	Use factors to describe the graph of a polynomial	**HSF-IF.C.7.c**

SET 42 • HSF-IF.C.7.a

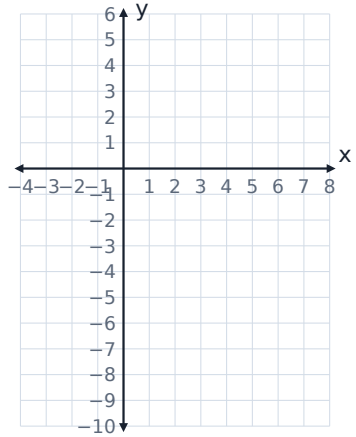
MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = x^2 - 4x - 5$.

Give the vertex, the x-intercepts, and the y-intercept.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

1279

QUESTION ID

S42-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A parabola opening up with vertex (2, −9), through (−1, 0), (0, −5), (4, −5), and (5, 0)

Written: Vertex (2, −9); x-intercepts (−1, 0) and (5, 0); y-intercept (0, −5)

GRADING NOTE Form required: all four points. Both the graph and the points are graded. $f(x) = (x - 5)(x + 1) = (x - 2)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the vertex and the axis of symmetry of $y = 2x^2 + 8x + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

1280

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

Vertex (−2, −5); axis x = −2

GRADING NOTE Form required: both. $x = -b \div (2a) = -8 \div 4 = -2$, and $y = 8 - 16 + 3 = -5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Does $y = -3x^2 + 6x - 1$ have a maximum or a minimum? Find it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

1281

QUESTION ID

S42-M-Q3

TYPE

Short answer

CORRECT ANSWER

A maximum of 2, at x = 1

GRADING NOTE Form required: maximum or minimum, its value, and where. $a = -3 < 0$, so the parabola opens down.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q3/5

1281

ANSWER KEY Reduced copy of the question page, with its answer

What are the x-intercepts of $y = (x + 3)(x - 1)$?

A. (3, 0) and (−1, 0)

B. (−3, 0) and (1, 0)

C. (−3, 0) and (−1, 0)

D. (0, −3) only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

1282

QUESTION ID

S42-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (−3, 0) and (1, 0)

GRADING NOTE $y = 0$ when $x + 3 = 0$ or $x - 1 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = x^2 + 4$ has no x-intercepts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

1283

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^2 + 4 \geq 4$ for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 1

Factor to find the zeros of a quadratic • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

ANSWER KEY Reduced copy of the question page, with its answer

Factor $x^2 - 4x - 5$, and give its zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

1285

QUESTION ID
S42-B1-Q1

TYPE
Short answer

CORRECT ANSWER
(x − 5)(x + 1); zeros 5 and −1

GRADING NOTE Form required: the factored form and both zeros.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the zeros of $2x^2 - 8$.

Zeros: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2 and −2

GRADING NOTE Form required: both zeros. $2(x - 2)(x + 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q2/5

1286

What are the zeros of $x^2 + 6x + 8$?

A. 2 and −4

B. −2 and −4

C. 2 and 4

D. −2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. −2 and −4

GRADING NOTE (x + 2)(x + 4).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q3/5

True or false?

The zeros of $(x - 7)(x + 2)$ are 7 and -2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each factor is 0 at one value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q4/5

True or false?

The zeros of $x^2 - 9x$ are 0 and −9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

1289

QUESTION ID
S42-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE x(x − 9): the zeros are 0 and 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

BACKWARD 2

Interpret key features of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

ANSWER KEY Reduced copy of the question page, with its answer

A table has x = −2, −1, 0, 1, 2 and y = 5, 2, 1, 2, 5.

Describe its symmetry and its minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Symmetric about x = 0; minimum 1 at x = 0.

GRADING NOTE Form required: both features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A profit function P(x) is positive for $10 < x < 50$, where x is the number of items sold.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The company makes a profit when it sells between 10 and 50 items.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q2/5

The highest point of a graph is (3, 40).

Which statement describes it?

A. A maximum of 40 at x = 3

B. A maximum of 3 at x = 40

C. A minimum of 3 at x = 40

D. A y-intercept of 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 40 at x = 3

GRADING NOTE The maximum value is the output, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q3/5

True or false?

A function is increasing on an interval when its outputs get larger as its inputs get larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE This is the definition.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q4/5

True or false?

The x-intercepts of a graph are where the output is greatest.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The output is 0 at an x-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 1

Relate a parabola's equation to its focus and directrix • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.2

Find the focus and directrix of $y = \frac{1}{8}x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

1297

QUESTION ID
S42-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Focus (0, 2); directrix y = −2

GRADING NOTE Form required: both. $x^2 = 8y = 4py$ with $p = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q1/5

Write an equation of the parabola with vertex (0, 0) and focus (0, −1).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

1298

QUESTION ID
S42-F1-Q2

TYPE
Short answer

CORRECT ANSWER
$x^2 = -4y$

GRADING NOTE Any equivalent form, such as $y = -\frac{1}{4}x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q2**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A parabola has focus (0, 3) and directrix y = −3.

Where is its vertex?

A. (0, 0)

B. (0, 3)

C. (0, −3)

D. (3, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

1299

QUESTION ID
S42-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. (0, 0)

GRADING NOTE Halfway between the focus and the directrix.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q3**
High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The parabola $x^2 = 12y$ opens upward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

1300

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $p = 3 > 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q4/5

True or false?

The parabola with focus (0, 1) and directrix $y = -1$ is $y = x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

1301

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x^2 = 4y$, so $y = \frac{1}{4}x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.2 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.C.7.a

FORWARD 2

Use factors to describe the graph of a polynomial • High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c

$f(x) = (x + 1)(x - 2)(x - 4)$.

Find the zeros and the y-intercept.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

1303

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Zeros −1, 2, and 4; y-intercept 8

GRADING NOTE Form required: all three zeros and the y-intercept. f(0) = (1)(−2)(−4) = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

At most how many turning points can the graph of a degree-3 polynomial have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

1304

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE Any equivalent form. At most n − 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q2/5

$f(x) = (x - 3)^2(x + 1)$.

What does the graph do at $x = 3$?

A. It touches the x-axis and turns around.

B. It is undefined.

C. It has a vertical asymptote.

D. It crosses the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

1305

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. It touches the x-axis and turns around.

GRADING NOTE The factor $(x - 3)$ is squared, so f does not change sign at 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q3/5

True or false?

$f(x) = x^3 - 8$ has exactly one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S42-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

1306

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x^3 = 8$ gives only $x = 2$ among the real numbers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q4**

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q4/5

True or false?

The y-intercept of $f(x) = (x - 1)(x - 2)(x - 3)$ is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

1307

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f(0) = (-1)(-2)(-3) = -6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b • INTERPRETING FUNCTIONS

Graph square root, cube root, absolute value, step, and piecewise-defined functions

MAIN	Algebra I main question	**HSF-IF.C.7.b**
BACKWARD 1	Find the domain and range of a function	**HSF-IF.A.1**
BACKWARD 2	Understand absolute value as distance from 0	**6.NS.C.7.c**
FORWARD 1	Solve radical equations and check for extraneous solutions	**HSA-REI.A.2**
FORWARD 2	Read values of inverses of power and root functions	**HSF-BF.B.4.c**

SET 43 • HSF-IF.C.7.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = |x - 2| - 3$.

Find the vertex of its graph and its x-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

1310

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Vertex (2, −3); x-intercepts (−1, 0) and (5, 0)

GRADING NOTE Form required: the vertex and both intercepts. $|x - 2| = 3$ gives $x = 5$ or $x = -1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q1/5

$f(x) = x + 1$ for $x < 2$, and $f(x) = 5 - x$ for $x \geq 2$.

Find f(0), f(2), and f(4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

1311

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

f(0) = 1, f(2) = 3, f(4) = 1

GRADING NOTE Form required: all three values. 2 is in the second piece.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

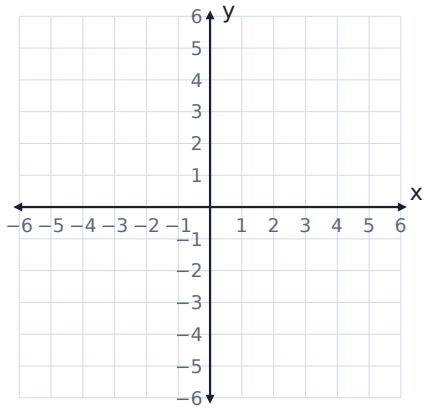
High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Graph f(x) = √(x + 4).

Give its domain and range.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

1312

QUESTION ID

S43-M-Q3

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve starting at (−4, 0) and rising to the right through (−3, 1), (0, 2), and (5, 3)

Written: Domain x ≥ −4; range y ≥ 0

GRADING NOTE Form required: the domain and the range. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which describes the graph of $y = \sqrt[3]{x}$?

A. It starts at (0, 0) and goes only to the right.

B. It is V-shaped with vertex (0, 0).

C. It has the horizontal asymptote y = 0.

D. It passes through (−8, −2), (0, 0), and (8, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

1313

QUESTION ID

S43-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. It passes through (−8, −2), (0, 0), and (8, 2).

GRADING NOTE Cube roots of negative numbers are negative, so the graph extends in both directions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q4/5

True or false?

For the greatest integer function, ⌊2.7⌋ = 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

1314

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE ⌊2.7⌋ is the greatest integer less than or equal to 2.7, which is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 1

Find the domain and range of a function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1

Over the real numbers, what is the domain of $f(x) = \sqrt{(x - 5)}$?

Domain: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$x \geq 5$

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the range of f(x) = |x|?

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

y ≥ 0

GRADING NOTE Form required: an inequality (or interval).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q2/5

Which value is NOT in the domain of $g(x) = \sqrt{(6 - x)}$?

A. 7

B. −3

C. 0

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE $6 - 7 = -1$ is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

Every real number is in the domain of $f(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Every real number has a real cube root.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

The range of $f(x) = x^2$ is all real numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The range is $y \geq 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

BACKWARD 2

Understand absolute value as distance from 0 • Grade 6 • The Number System • 6.NS.C.7.c

ANSWER KEY Reduced copy of the question page, with its answer

What is |−9|?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

GRADING NOTE Any equivalent form. −9 is 9 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Which is farther from 0, −7 or 5? Explain with absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−7, because |−7| = 7 and |5| = 5.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which value equals |3 − 8|?

A. 5

B. −5

C. 11

D. −11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

GRADING NOTE |−5| = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|-4| = 4$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE −4 is 4 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$|0| = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 0 is 0 units from 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 1

Solve radical equations and check for extraneous solutions • High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Solve √(x − 2) = 4.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

1328

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any equivalent form. x − 2 = 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q1**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q1/5

Solve $\sqrt{(x + 6)} = x$. Name any extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

1329

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$x = 3$; $x = -2$ is extraneous.

GRADING NOTE Form required: the solution and the extraneous value. $x^2 - x - 6 = 0$ gives 3 and -2, but $\sqrt{4} = 2 \neq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q2/5

Solve $\sqrt[3]{(x + 1)} = 2$.

A. x = 7

B. x = 3

C. x = 9

D. x = 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

1330

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 7

GRADING NOTE x + 1 = 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q3/5

True or false?

$\sqrt{x} + 3 = 0$ has no real solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

1331

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $\sqrt{x} \geq 0$, so $\sqrt{x} + 3 \geq 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q4/5

True or false?

$x = 16$ is a solution of $\sqrt{x} = -4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

1332

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\sqrt{16} = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q5**

High School — Algebra II (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.C.7.b

FORWARD 2

Read values of inverses of power and root functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = x^3$.

Find $g^{-1}(27)$.

$g^{-1}(27) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

1334

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. $3^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The point (4, 2) is on the graph of $f(x) = \sqrt{x}$.

Which point must be on the graph of f^{-1}?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

1335

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 4)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = x^2$ with domain $x \geq 0$.

What is $f^{-1}(49)$?

A. -7

B. 24.5

C. 2,401

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

1336

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

GRADING NOTE $f(7) = 49$, and 7 is in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The inverse of $f(x) = x^3$ is $f^{-1}(x) = \sqrt[3]{x}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

1337

QUESTION ID
S43-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Cubing and taking the cube root undo each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q4/5

True or false?

If (3, 10) is on the graph of f, then (3, 10) is on the graph of f^{-1}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

1338

QUESTION ID
S43-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE (10, 3) is on the graph of f^{-1}.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.c • Set 43 • FORWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e • INTERPRETING FUNCTIONS

Graph exponential functions and show intercepts and end behavior

MAIN	Algebra I main question	**HSF-IF.C.7.e**
BACKWARD 1	Evaluate integer exponents	**8.EE.A.1**
BACKWARD 2	Evaluate exponential functions	**HSF-IF.A.2**
FORWARD 1	Solve exponential equations with logarithms	**HSF-LE.A.4**
FORWARD 2	Understand logarithms as inverses of exponential functions	**HSF-BF.B.5**

SET 44 • HSF-IF.C.7.e

MAIN

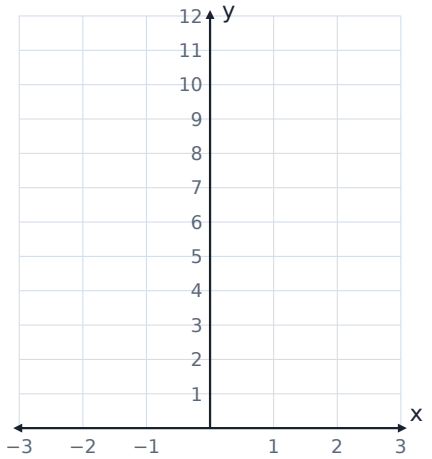
Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

ANSWER KEY Reduced copy of the question page, with its answer

Graph $f(x) = 3 \cdot 2^x$ for $-2 \le x \le 2$.

Give its y-intercept and its horizontal asymptote.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

1341

QUESTION ID

S44-M-Q1

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A curve through (−2, 0.75), (−1, 1.5), (0, 3), (1, 6), and (2, 12), rising more steeply to the right

Written: y-intercept (0, 3); asymptote y = 0

GRADING NOTE Form required: both. Both the graph and the answer are graded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q1/5

$g(x) = 50(0.8)^x$.

Is this growth or decay? Give its y-intercept, and say what happens to g(x) as x gets large.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

1342

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

Decay; y-intercept 50; g(x) gets closer and closer to 0.

GRADING NOTE Form required: all three parts. 0.8 < 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q2/5

$h(x) = 2^x - 4$.

Find the y-intercept, the x-intercept, and the horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

1343

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

y-intercept (0, −3);
x-intercept (2, 0);
asymptote y = −4

GRADING NOTE Form required: all three. $2^0 - 4 = -3$, and $2^x = 4$ at $x = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which function's graph rises from left to right and passes through (0, 5)?

A. $y = 1.2(5)^x$

B. $y = 0.8(5)^x$

C. $y = 5(0.8)^x$

D. $y = 5(1.2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

1344

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 5(1.2)^x$

GRADING NOTE A base greater than 1 gives growth, and the coefficient is the y-intercept.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph of $y = 4(0.5)^x$ crosses the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

1345

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4(0.5)^x$ is always positive; the x-axis is an asymptote.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 44 • MAIN • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 1

Evaluate integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate 2^{-2}.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

GRADING NOTE Any equivalent form, such as 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $3 \cdot 2^{-1}$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE Any equivalent form, such as 3/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which number equals $(0.5)^{-1}$?

A. -0.5

B. 0.5

C. 2

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2

GRADING NOTE $1 \div 0.5 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$5^0 = 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Any nonzero number to the 0 power is 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$4^{-1} = \frac{1}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^{-1} = 1 \div 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

BACKWARD 2

Evaluate exponential functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 50(0.8)^x$.

Find f(2).

f(2) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

32

GRADING NOTE Any equivalent form. 50(0.64).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$g(x) = 2^x - 4$.

Find g(3).

g(3) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. 8 − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q2/5

$h(x) = 3 \cdot 2^x$.

Find $h(-1)$.

A. -1.5

B. 6

C. -6

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 1.5

GRADING NOTE $3 \cdot \frac{1}{2} = 1.5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q3/5

True or false?

If $f(x) = 10^x$, then $f(3) = 1{,}000$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^3 = 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S44-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q4/5

True or false?

If $f(x) = 3^x$, then $f(2) = 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $3^2 = 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 44 • BACKWARD 2 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 1

Solve exponential equations with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve $3 \cdot 2^x = 48$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

1359

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Any equivalent form. $2^x = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $50(0.8)^x = 20$. Round to the nearest tenth.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

1360

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4.1

GRADING NOTE Form required: rounded to the nearest tenth. $x = \log 0.4 \div \log 0.8 \approx 4.11$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^x = 12$.

A. x = log 6

B. x = log 12 ÷ log 2 ≈ 3.58

C. x = 3.5

D. x = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

1361

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x = log 12 ÷ log 2 ≈ 3.58

GRADING NOTE Take the logarithm of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

log 1 = 0

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

1362

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q4**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

log 100 = 2

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

1363

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $10^2 = 100$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • HSF-IF.C.7.e

FORWARD 2

Understand logarithms as inverses of exponential functions • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

Write the inverse of $f(x) = 2^x$.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

1365

QUESTION ID
S44-F2-Q1

TYPE
Short answer

CORRECT ANSWER
$\log_2 x$

GRADING NOTE Any equivalent form, such as log x ÷ log 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Explain why $\log_3 81 = 4$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

1366

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3^4 = 81$, and $\log_3 81$ is the exponent on 3 that gives 81.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph of $y = \log_2 x$ is the reflection of the graph of $y = 2^x$ across which line?

A. $x = 0$

B. $y = 0$

C. $y = x$

D. $y = -x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

1367

QUESTION ID
S44-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. y = x

GRADING NOTE Inverse functions reflect across y = x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q3/5

True or false?

The domain of $y = \log x$ is $x > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

1368

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the range of $y = 10^x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q4/5

True or false?

$\log_2(2^5) = 2$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

1369

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log_2(2^5) = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.5 • Set 44 • FORWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a • INTERPRETING FUNCTIONS

Factor and complete the square to show zeros, extreme values, and symmetry, and interpret them

MAIN	Algebra I main question	**HSF-IF.C.8.a**
BACKWARD 1	Complete the square	**HSA-SSE.B.3.b**
BACKWARD 2	Use the greatest common factor	**6.NS.B.4**
FORWARD 1	Prove polynomial identities	**HSA-APR.C.4**
FORWARD 2	Complete the square to find the center of a hyperbola	**HSG-GPE.A.3**

SET 45 • HSF-IF.C.8.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a

$f(x) = x^2 - 8x + 12$.

Write f in factored form and in vertex form. Give the zeros, the vertex, and the axis of symmetry.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

1372

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

(x − 2)(x − 6) and (x − 4)² − 4; zeros 2 and 6; vertex (4, −4); axis x = 4

GRADING NOTE Form required: both forms and all three features.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A company's profit is $P(x) = -2x^2 + 120x - 1{,}000$ dollars when it charges x dollars.

Complete the square to find the price that gives the most profit, and the greatest profit.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

1373

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

$P(x) = -2(x - 30)^2 + 800$; a price of $30 gives the greatest profit, $800.

GRADING NOTE Form required: the completed square, the price, and the profit. $-2(x^2 - 60x + 900) + 1{,}800 - 1{,}000$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A rock's height is $h(t) = -16t^2 + 32t + 48$ feet after t seconds.

Factor to find when it hits the ground.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

1374

QUESTION ID

S45-M-Q3

TYPE

Short answer

CORRECT ANSWER

h(t) = −16(t − 3)(t + 1); at t = 3 seconds

GRADING NOTE Form required: the factored form and the time. t = −1 is not in the domain.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Which form of $f(x) = x^2 + 6x + 5$ shows the vertex directly?

A. $(x + 1)(x + 5)$

B. $x^2 + 6x + 5$

C. $(x + 3)^2 - 4$

D. $x(x + 6) + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

1375

QUESTION ID

S45-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(x + 3)^2 - 4$

GRADING NOTE Vertex form $a(x - h)^2 + k$ shows the vertex $(-3, -4)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q4/5

True or false?

The axis of symmetry of $y = (x - 1)(x - 7)$ is $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

1376

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The axis is halfway between the zeros 1 and 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 45 • MAIN • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 1

Complete the square • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b

Complete the square.

$x^2 - 8x + 12$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x - 4)^2 - 4$

GRADING NOTE Form required: $(x - h)^2 + k$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square.

$x^2 + 3x$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1.5)^2 - 2.25$

GRADING NOTE Form required: $(x - h)^2 + k$, with fractions or decimals.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q2/5

Which expression equals $-(x^2 - 6x)$?

A. $-(x - 3)^2 - 9$

B. $-(x - 3)^2 + 9$

C. $(x - 3)^2 - 9$

D. $-(x + 3)^2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $-(x - 3)^2 + 9$

GRADING NOTE $x^2 - 6x = (x - 3)^2 - 9$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 2x + 10 = (x + 1)^2 + 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x + 1)^2 = x^2 + 2x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x - 5)^2 - 3$ has its minimum at $x = -5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

1382

QUESTION ID
S45-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its minimum is at $x = 5$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.b • Set 45 • BACKWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

BACKWARD 2

Use the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Use the greatest common factor to write 16 + 40 as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8(2 + 5)

GRADING NOTE Form required: the GCF times a sum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the greatest common factor of 32 and 48.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q2/5

Which expression equals 36 + 27 and uses the greatest common factor?

A. 9(4 + 3)

B. 3(12 + 6)

C. 6(6 + 27)

D. 9(4 + 27)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 9(4 + 3)

GRADING NOTE $9 \cdot 4 = 36$ and $9 \cdot 3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q3/5

True or false?

The greatest common factor of 12 and 18 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 6 divides both, and no larger number does.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$20 + 50 = 5(4 + 5)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $5(4 + 5) = 45$. $20 + 50 = 10(2 + 5)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 1

Prove polynomial identities • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

ANSWER KEY Reduced copy of the question page, with its answer

Show that $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ by expanding both sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

1390

QUESTION ID
S45-F1-Q1

TYPE
Short answer

CORRECT ANSWER
Left: $x^4 + 2x^2y^2 + y^4$. Right: $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The two sides are equal.

GRADING NOTE Form required: both sides expanded.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $x = 2$ and $y = 1$ in $x^2 - y^2$, $2xy$, and $x^2 + y^2$ to make a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

1391

QUESTION ID
S45-F1-Q2

TYPE
Short answer

CORRECT ANSWER
3, 4, 5

GRADING NOTE Any order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q2/5

Which equation is an identity (true for all numbers a and b)?

A. $a^3 + b^3 = (a + b)^3$

B. $a^3 - b^3 = (a - b)^3$

C. $a^2 + b^2 = (a + b)^2$

D. $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

1392

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D.

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

GRADING NOTE Expanding $(a - b)(a^2 + ab + b^2)$ gives $a^3 - b^3$. Each other choice fails for $a = 2$, $b = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q3/5

True or false?

$(a + b)^2 - (a - b)^2 = 4ab$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

1393

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$(x + 1)^3 = x^3 + 1$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

1394

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 1)^3 = x^3 + 3x^2 + 3x + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.4 • Set 45 • FORWARD 1 • Q5/5

SET 45 • HSF-IF.C.8.a

FORWARD 2

Complete the square to find the center of a hyperbola • High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.3

ANSWER KEY Reduced copy of the question page, with its answer

Complete the square to find the center of the hyperbola $x^2 - 4y^2 - 6x - 8y - 11 = 0$.

Center: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

1396

QUESTION ID
S45-F2-Q1

TYPE
Short answer

CORRECT ANSWER
(3, −1)

GRADING NOTE Any equivalent form. $(x - 3)^2 - 4(y + 1)^2 = 11 + 9 - 4 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Where are the vertices of the hyperbola $x^2 \div 9 - y^2 \div 16 = 1$?

Vertices: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

1397

QUESTION ID
S45-F2-Q2

TYPE
Short answer

CORRECT ANSWER
(3, 0) and (−3, 0)

GRADING NOTE Form required: both vertices.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which equation is a hyperbola?

A. $y = x^2 \div 4$

B. $x^2 \div 4 + y^2 \div 9 = 1$

C. $x^2 \div 4 - y^2 \div 9 = 1$

D. $x^2 + y^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

1398

QUESTION ID
S45-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
C. $x^2 \div 4 - y^2 \div 9 = 1$

GRADING NOTE A hyperbola has a difference of squared terms; the others are an ellipse, a parabola, and a circle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q3/5

True or false?

For a hyperbola, the difference of the distances from any of its points to the two foci is constant.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

1399

QUESTION ID
S45-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE This is the definition of a hyperbola.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q4/5

True or false?

The hyperbola $x^2 - y^2 = 1$ passes through (0, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

1400

QUESTION ID
S45-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $0 - 1 = -1$, not 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5
High School — Precalculus (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.3 • Set 45 • FORWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b • INTERPRETING FUNCTIONS

Use the properties of exponents to interpret growth and decay rates

MAIN	Algebra I main question	**HSF-IF.C.8.b**
BACKWARD 1	Read growth and decay from an exponential function	**HSF-IF.C.7.e**
BACKWARD 2	Solve percent increase and decrease problems	**7.RP.A.3**
FORWARD 1	Sum a geometric series in a savings context	**HSA-SSE.B.4**
FORWARD 2	Find times in exponential models with logarithms	**HSF-LE.A.4**

SET 46 • HSF-IF.C.8.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b

ANSWER KEY Reduced copy of the question page, with its answer

$y = (1.02)^t$, where t is in years.

Is this growth or decay, and what is the percent rate of change per year?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

1403

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

Growth, at 2% per year

GRADING NOTE Form required: growth or decay and the percent. 1.02 = 1 + 0.02.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (1.01)^{12t}$, where t is in years.

Rewrite it to find the yearly growth factor (to four decimal places) and the yearly percent rate.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

1404

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = (1.01^{12})^{t} \approx (1.1268)^{t}$; about 12.68% per year

GRADING NOTE Form required: the yearly factor and the percent. $(1.01)^{12t} = ((1.01)^{12})^{t}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q2/5

$y = 200(0.97)^t$ gives an amount after t years.

What is the yearly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

1405

QUESTION ID

S46-M-Q3

TYPE

Short answer

CORRECT ANSWER

3% per year

GRADING NOTE Any equivalent form. 0.97 = 1 − 0.03.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$y = (1.2)^{t/10}$, where t is in years.

By what percent does y grow every 10 years?

A. 2%

B. 12%

C. 1.2%

D. 20%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

1406

QUESTION ID

S46-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20%

GRADING NOTE When t increases by 10, the exponent increases by 1, so y is multiplied by 1.2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q4/5

True or false?

$y = 100(0.5)^{t/8}$ halves every 8 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

1407

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Each increase of 8 in t multiplies y by 0.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 46 • MAIN • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 1

Read growth and decay from an exponential function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e

Is $y = 30(1.5)^x$ increasing or decreasing? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Increasing, because the base 1.5 is greater than 1.

GRADING NOTE Form required: the answer and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is the y-intercept of $y = 7(0.9)^x$?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form, such as (0, 7).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q2/5

Which function is decreasing?

A. $y = 2(1.6)^x$

B. $y = 0.6(1.2)^x$

C. $y = 2(0.6)^x$

D. $y = 0.6(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2(0.6)^x$

GRADING NOTE Only the first has a base between 0 and 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q3/5

True or false?

The graph of $y = 3^x$ has the horizontal asymptote $y = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3^x approaches 0 as x decreases.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q4/5

True or false?

$y = 5(1.1)^x$ has y-intercept 1.1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE At x = 0, y = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 46 • BACKWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

BACKWARD 2

Solve percent increase and decrease problems • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

A $40 shirt is marked up 15%.

What is the new price?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$46

GRADING NOTE Any equivalent form. 40 × 1.15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A town grows from 2,000 to 2,300 people.

What is the percent increase?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15%

GRADING NOTE Any equivalent form. 300 ÷ 2,000.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q2/5

A price drops from $80 to $60.

What is the percent decrease?

A. 25%

B. 20%

C. 33%

D. 75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 25%

GRADING NOTE 20 ÷ 80 = 0.25.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 1.06 increases a number by 6%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 1.06 = 1 + 0.06.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Multiplying by 0.85 decreases a number by 85%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases the number by 15%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 1

Sum a geometric series in a savings context • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

You save $100 the first month and 2% more each month after that.

Write the total for 12 months as a sum formula, and find it to the nearest dollar.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

1421

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$100(1.02^{12} - 1) \div 0.02 \approx \$1{,}341$

GRADING NOTE Form required: the formula and the rounded total.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q1/5

Find $4 + 4(1.5) + 4(1.5)^2 + 4(1.5)^3$.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

1422

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

32.5

GRADING NOTE Any equivalent form. 4 + 6 + 9 + 13.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which expression is the sum of the first n terms of $a + ar + ar^2 + \ldots$ $(r \neq 1)$?

A. $a \div (1 - r)$

B. $a(1 - r) \div (1 - r^n)$

C. $a(1 - r^n) \div (1 - r)$

D. nar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

1423

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $a(1 - r^n) \div (1 - r)$

GRADING NOTE The finite geometric series formula.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1 + 0.5 + 0.25 + 0.125 = 1.875$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

1424

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Add the terms.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q4/5

True or false?

The sum of 10 terms of $3 + 3 + 3 + \ldots$ is 3^{10}.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

1425

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $10 \cdot 3 = 30$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 46 • FORWARD 1 • Q5/5

SET 46 • HSF-IF.C.8.b

FORWARD 2

Find times in exponential models with logarithms • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$y = 100(0.5)^{t/8}$.

Find t when y = 10. Round to the nearest tenth.

t ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

1427

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26.6

GRADING NOTE Form required: rounded to the nearest tenth. t = 8 · log 0.1 ÷ log 0.5 ≈ 26.58.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the continuous rate r for which $e^{10r} = 2$. Round to the nearest thousandth.

r ≈ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

1428

QUESTION ID
S46-F2-Q2

TYPE
Short answer

CORRECT ANSWER
0.069

GRADING NOTE Form required: rounded to the nearest thousandth. r = ln 2 ÷ 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q2/5

To the nearest tenth, how long does $y = (1.05)^t$ take to double?

A. 20

B. 7.2

C. 10.5

D. 14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

1429

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 14.2

GRADING NOTE $t = \log 2 \div \log 1.05 \approx 14.21$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

If $y = 2^{t/3}$, then y doubles every 3 units of t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

1430

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Adding 3 to t adds 1 to the exponent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q4/5

True or false?

$\log(2^{10}) = 20$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

1431

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $\log(2^{10}) = 10 \log 2 \approx 3.01$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 46 • FORWARD 2 • Q5/5

SET 47 • HSF-IF.C.9 • INTERPRETING FUNCTIONS

Compare properties of two functions represented in different ways

MAIN	Algebra I main question	**HSF-IF.C.9**
BACKWARD 1	Interpret key features in context	**HSF-IF.B.4**
BACKWARD 2	Compute average rates of change	**HSF-IF.B.6**
FORWARD 1	Compare periodic functions given in different forms	**HSF-TF.B.5**
FORWARD 2	Compare normal distributions by mean and standard deviation	**HSS-ID.A.4**

SET 47 • HSF-IF.C.9

MAIN

Algebra I main question • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9

f(x) = 3x + 2. The table shows g.

Which function has the greater value at x = 3? Which has the greater average rate of change from x = 0 to x = 3?

x	0	1	2	3
g(x)	1	4	9	16

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

1434

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

g at x = 3 (16 compared with 11); g has the greater rate (5 compared with 3).

GRADING NOTE Form required: both comparisons with values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q1/5

Function A is $y = -2(x - 3)^2 + 10$. Function B is a parabola that opens down with vertex (1, 12).

Which has the greater maximum value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

1435

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

B: its maximum is 12, and A's is 10.

GRADING NOTE Form required: the function and both maxima.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q2/5

Function f has y-intercept 4 and increases by 5 for each 1-unit increase in x. g(x) = 5x − 1.

Compare their slopes and y-intercepts.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

1436

QUESTION ID

S47-M-Q3

TYPE

Short answer

CORRECT ANSWER

Same slope, 5; f has the greater y-intercept (4 compared with −1), so their graphs are parallel lines.

GRADING NOTE Form required: both comparisons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 1.08^x$ grows 8% per year. The table shows Q.

Which grows faster?

x	0	1	2
Q(x)	1	1.1	1.21

A. P: its graph is a line.

B. Q: it grows 10% per year.

C. P: it grows 8% per year and Q grows 1%.

D. They grow at the same rate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

1437

QUESTION ID

S47-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Q: it grows 10% per year.

GRADING NOTE Q is multiplied by 1.1 each year: 1, 1.1, 1.21.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q4/5

True or false?

$f(x) = x^2$, and g is a parabola opening up with vertex $(0, -4)$. The two functions have the same minimum value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

1438

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum of f is 0 and the minimum of g is −4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.9 • Set 47 • MAIN • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 1

Interpret key features in context • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

T(h) is the temperature h hours after midnight, and its minimum is T(6) = 4.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The lowest temperature, 4 degrees, happened 6 hours after midnight (6 a.m.).

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A graph of savings S(m) is horizontal from m = 3 to m = 5.

What does that mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

The savings stayed the same from month 3 to month 5.

GRADING NOTE Any equivalent wording.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q2/5

A function's graph is an unbroken curve. It is positive for $0 < x < 8$ and negative for $x > 8$.

What is true at $x = 8$?

A. The output is greatest.

B. The output is 8.

C. The output is 0.

D. The function is undefined.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The output is 0.

GRADING NOTE An unbroken curve that changes sign passes through the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S47-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

A y-intercept gives the starting value of a quantity that changes over time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It is the output at time 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

An x-intercept tells the starting value of a quantity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE An x-intercept is where the output is 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

BACKWARD 2

Compute average rates of change • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2^x$.

Find the average rate of change from $x = 1$ to $x = 3$.

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

GRADING NOTE Any equivalent form. (8 − 2) ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A savings account had $100 at week 0 and $180 at week 4.

What is the average rate of change?

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$20 per week

GRADING NOTE Any equivalent form, with units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q2/5

$g(x) = x^2 - 1$.

What is the average rate of change from $x = -1$ to $x = 2$?

A. 1

B. −1

C. 3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1

GRADING NOTE $(3 - 0) \div 3 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q3/5

True or false?

The average rate of change of y = 5 is 0 on every interval.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The output never changes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q4/5

True or false?

The average rate of change of $y = x^2$ from $x = 0$ to $x = 2$ is 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(4 - 0) \div 2 = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.6 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 1

Compare periodic functions given in different forms • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

f(x) = 3sin(2x). Function g has amplitude 4 and period π.

Which has the greater amplitude? Do they have the same period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

1452

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

g has the greater amplitude (4 compared with 3); yes, both periods are π.

GRADING NOTE Form required: both parts. The period of f is 2π ÷ 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Tide A rises to 12 ft and falls to 2 ft. Tide B is $h(t) = 6 + 4\sin t$.

Which has the greater amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

1453

QUESTION ID
S47-F1-Q2

TYPE
Short answer

CORRECT ANSWER
Tide A: 5 ft compared with 4 ft.

GRADING NOTE Form required: the tide and both amplitudes. (12 − 2) ÷ 2 = 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q2/5

What is the period of y = sin(4x)?

A. π ÷ 2

B. π ÷ 4

C. 2π

D. 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

1454

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. π ÷ 2

GRADING NOTE 2π ÷ 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q3/5

True or false?

y = 2cos x and y = 2sin x have the same amplitude and the same period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

1455

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Both have amplitude 2 and period 2π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q4/5

True or false?

y = 5sin x has a maximum value of 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

1456

QUESTION ID
S47-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its maximum value is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 47 • FORWARD 1 • Q5/5

SET 47 • HSF-IF.C.9

FORWARD 2

Compare normal distributions by mean and standard deviation • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Test A scores are normal with mean 70 and standard deviation 5. Test B scores are normal with mean 70 and standard deviation 10.

Which test's scores are more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5

1458

QUESTION ID
S47-F2-Q1

TYPE
Short answer

CORRECT ANSWER
Test B: its standard deviation is larger.

GRADING NOTE Form required: the test and the reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Heights are normal with mean 64 in and standard deviation 3 in.

About what percent are between 61 in and 67 in?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5

1459

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 68%

GRADING NOTE Any equivalent form. That interval is within 1 standard deviation of the mean.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q2**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q2/5

Scores are normal with mean 100 and standard deviation 15.

About what percent of scores are above 130?

A. About 16%

B. About 34%

C. About 2.5%

D. About 5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5

1460

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. About 2.5%

GRADING NOTE 130 is 2 standard deviations above the mean; about 95% lie within 2, so about 2.5% lie above.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q3**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q3/5

True or false?

A normal distribution is symmetric about its mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5

1461

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Its curve is a symmetric bell shape.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q4/5

True or false?

In a normal distribution, about 95% of the data lie within 1 standard deviation of the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5

1462

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE About 68% lie within 1; about 95% lie within 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.4 • Set 47 • FORWARD 2 • Q5/5

Write an explicit expression or a recursive process for a function from a context

MAIN	Algebra I main question	**HSF-BF.A.1.a**
BACKWARD 1	Evaluate a sequence from a recursive or explicit rule	**HSF-IF.A.3**
BACKWARD 2	Write equations in two or more variables	**HSA-CED.A.2**
FORWARD 1	Build a composite function from a context	**HSF-BF.A.1.c**
FORWARD 2	Find and interpret the inverse of a model	**HSF-BF.B.4.b**

SET 48 • HSF-BF.A.1.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a

ANSWER KEY Reduced copy of the question page, with its answer

A phone plan has a $60 activation fee and costs $35 per month.

Write an explicit expression for the total cost C(n) after n months, and a recursive rule.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

1465

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

C(n) = 60 + 35n; C(0) = 60 and C(n) = C(n − 1) + 35 for n ≥ 1

GRADING NOTE Form required: both the explicit expression and the recursive rule.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q1/5

A tree is 4 ft tall and grows 1.5 ft per year.

Write a function for its height h(t) after t years.

h(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

1466

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 + 1.5t

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

A ball is dropped from 10 m. Each bounce reaches 60% of the height of the bounce before.

Write a function for the height h(n) of the nth bounce.

h(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

1467

QUESTION ID

S48-M-Q3

TYPE

Short answer

CORRECT ANSWER

$10(0.6)^n$

GRADING NOTE Any equivalent form. The first bounce reaches 10(0.6) = 6 m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q3/5

A table shows n = 1, 2, 3, 4 and a(n) = 7, 11, 15, 19.

Which recursive process fits?

A. a(1) = 4, a(n) = a(n − 1) + 7

B. a(1) = 7, a(n) = 4a(n − 1)

C. a(1) = 7, a(n) = a(n − 1) + 4

D. a(n) = 7n + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

1468

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. a(1) = 7, a(n) = a(n − 1) + 4

GRADING NOTE Start at 7 and add 4 each time. (7n + 4 gives 11 for n = 1.)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q4/5

True or false?

The steps "start at 500 and multiply by 1.03 each year" give f(t) = 500 + 1.03t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

1469

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Repeated multiplication gives $f(t) = 500(1.03)^t$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.a • Set 48 • MAIN • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 1

Evaluate a sequence from a recursive or explicit rule • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 2$ and $f(n) = 3f(n - 1)$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

54

GRADING NOTE Any equivalent form. 2, 6, 18, 54.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q1/5

$f(n) = 10 - 2n$ for $n \geq 1$.

List the first four terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8, 6, 4, 2

GRADING NOTE Form required: the four terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q2/5

$f(1) = 5$ and $f(n) = f(n - 1) - 3$.

What is $f(3)$?

A. -1

B. 11

C. 2

D. -4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −1

GRADING NOTE 5, 2, −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q3/5

True or false?

The terms f(1), f(2), f(3) of a sequence are its outputs for the inputs 1, 2, 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE A sequence is a function on the integers.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q4/5

True or false?

$f(n) = n^2$ gives the terms 1, 2, 3, 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It gives 1, 4, 9, 16.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

BACKWARD 2

Write equations in two or more variables • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2

ANSWER KEY Reduced copy of the question page, with its answer

A car rental costs $45 per day plus $0.20 per mile.

Write an equation for the cost C for d days and m miles.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

C = 45d + 0.20m

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an equation for the perimeter P of a rectangle with length l and width w.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

P = 2l + 2w

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q2/5

A $900 laptop loses $150 of value each year.

Which equation gives its value V after t years?

A. $V = 900 - 150t$

B. $V = 150 - 900t$

C. $V = 900(150)^t$

D. $V = 900 + 150t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. V = 900 − 150t

GRADING NOTE Start at 900 and subtract 150 per year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q3/5

True or false?

$A = s^2$ gives the area of a square with side s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Side times side.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q4/5

True or false?

y = 12x gives the number of feet in x inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE y = 12x gives the number of inches in x feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.2 • Set 48 • BACKWARD 2 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 1

Build a composite function from a context • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c

An oil spill's radius is r(t) = 2t meters after t hours, and the area of a circle is A(r) = πr².

Write A(r(t)) and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

1483

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A(r(t)) = 4πt²: the area of the spill after t hours.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A store takes 20% off and then $5 off: d(x) = 0.8x and c(x) = x − 5.

Find c(d(50)).

c(d(50)) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

1484

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

35

GRADING NOTE Any equivalent form. d(50) = 40, and c(40) = 35.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q2/5

With d(x) = 0.8x and c(x) = x − 5, which expression is d(c(x))?

A. 0.8x − 4

B. 0.8x + 4

C. 0.8x − 5

D. x − 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

1485

QUESTION ID
S48-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
A. 0.8x − 4

GRADING NOTE 0.8(x − 5) = 0.8x − 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q3/5

True or false?

With these functions, c(d(50)) and d(c(50)) are different.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

1486

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE c(d(50)) = 35 and d(c(50)) = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q4/5

True or false?

If f(x) = 3x and g(x) = x + 2, then f(g(x)) = 3x + 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

1487

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE f(g(x)) = 3(x + 2) = 3x + 6.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.A.1.c • Set 48 • FORWARD 1 • Q5/5

SET 48 • HSF-BF.A.1.a

FORWARD 2

Find and interpret the inverse of a model • High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b

C(n) = 60 + 35n is the total cost after n months.

Find the inverse and say what it gives.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

1489

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

n = (C − 60) ÷ 35: the number of months for a total cost of C dollars.

GRADING NOTE Any equivalent form; the meaning is required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q1/5

Show that f(x) = 5x − 2 and g(x) = (x + 2) ÷ 5 are inverses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

1490

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

f(g(x)) = (x + 2) − 2 = x and g(f(x)) = (5x − 2 + 2) ÷ 5 = x.

GRADING NOTE Form required: both compositions.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

F = 1.8C + 32 converts Celsius to Fahrenheit.

Which formula is its inverse?

A. C = F ÷ 1.8 − 32

B. C = 1.8F + 32

C. C = (F + 32) ÷ 1.8

D. C = (F − 32) ÷ 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

1491

QUESTION ID
S48-F2-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. C = (F − 32) ÷ 1.8

GRADING NOTE Subtract 32, then divide by 1.8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q3/5

True or false?

If $f(2) = 9$, then $f^{-1}(9) = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

1492

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The inverse reverses each pair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q4/5

True or false?

The inverse of $f(x) = 2x$ is $f^{-1}(x) = -2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

1493

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $f^{-1}(x) = x \div 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Building Functions • HSF-BF.B.4.b • Set 48 • FORWARD 2 • Q5/5

Combine standard function types using arithmetic operations

MAIN	Algebra I main question	**HSF-BF.A.1.b**
BACKWARD 1	Add, subtract, and multiply polynomials	**HSA-APR.A.1**
BACKWARD 2	Evaluate and interpret functions	**HSF-IF.A.2**
FORWARD 1	Rewrite rational expressions such as average cost	**HSA-APR.D.6**
FORWARD 2	Graph and interpret an average-cost rational function	**HSF-IF.C.7.d**

SET 49 • HSF-BF.A.1.b

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Revenue is R(x) = 25x and cost is C(x) = 400 + 9x dollars for x items.

Write the profit function P(x) in simplest form and find P(50).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

1496

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

P(x) = 16x − 400; P(50) = 400

GRADING NOTE Form required: P(x) simplified and the value. P = R − C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q1/5

A ball's height is $h(t) = -4.9t^2 + 30t + 2$ meters after t seconds.

Name the three simple functions added together, and what each one describes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

1497

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

The constant 2 (the starting height), the linear 30t (height gained from the upward speed), and the quadratic $-4.9t^2$ (the pull of gravity).

GRADING NOTE Any equivalent description of the three parts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q2/5

$f(x) = x^2 + 1$ and $g(x) = 2x - 3$.

Find $(f - g)(x)$ in simplest form, and find $(f \cdot g)(2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

1498

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(f - g)(x) = x^2 - 2x + 4$; $(f \cdot g)(2) = 5$

GRADING NOTE Form required: the simplified difference and the value. $f(2) = 5$ and $g(2) = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q3/5

The temperature of tea is $T(t) = 20 + 70(0.9)^t$ °C after t minutes.

Which description fits this combination of functions?

A. A linear function with slope 0.9

B. An exponential function that grows 90% per minute

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

D. A constant temperature of 90 °C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

1499

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. A constant room temperature of 20 °C plus a difference of $70(0.9)^t$ that decays

GRADING NOTE The constant 20 is added to an exponential decay term that starts at 70.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q4/5

True or false?

If $f(x) = 3x$ and $g(x) = x^2$, then $(f + g)(2) = 10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

1500

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $f(2) + g(2) = 6 + 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.1.b • Set 49 • MAIN • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 1

Add, subtract, and multiply polynomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Subtract.

$25x - (400 + 9x)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16x − 400

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$(x^2 + 1) + (2x - 3)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$x^2 + 2x - 2$

GRADING NOTE Form required: simplified.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q2/5

Multiply.

$(x^2 + 1)(2x - 3)$

A. $2x^3 + 2x - 3$

B. $2x^2 + 2x - 3$

C. $2x^3 - 3x^2 - 3$

D. $2x^3 - 3x^2 + 2x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $2x^3 - 3x^2 + 2x - 3$

GRADING NOTE Multiply each term of the first by each term of the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q3/5

True or false?

The product of two polynomials is a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Polynomials are closed under multiplication.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q4/5

True or false?

$(x + 2) - (x - 2) = 0$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $(x + 2) - (x - 2) = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

BACKWARD 2

Evaluate and interpret functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

$P(x) = 16x - 400.$

Find x so that $P(x) = 0$.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

1508

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

25

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$T(t) = 20 + 70(0.9)^t$.

Find T(0).

T(0) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

1509

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

90

GRADING NOTE Any equivalent form. $(0.9)^0 = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

R(x) is the revenue in dollars from selling x items.

What does R(10) = 250 mean?

A. Selling 250 items brings in $10.

B. Each item costs $10.

C. The profit is $250.

D. Selling 10 items brings in $250.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

1510

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Selling 10 items brings in $250.

GRADING NOTE Input: items. Output: dollars of revenue.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q3/5

True or false?

If C(x) = 400 + 9x, then C(0) = 400.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

1511

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $9 \cdot 0 = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q4/5

True or false?

If $h(t) = -4.9t^2 + 30t + 2$, then $h(0) = 30$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

1512

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $h(0) = 2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 1

Rewrite rational expressions such as average cost • High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

ANSWER KEY Reduced copy of the question page, with its answer

Write the average cost A(x) = (400 + 9x) ÷ x as a constant plus a fraction.

A(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

1514

QUESTION ID
S49-F1-Q1

TYPE
Short answer

CORRECT ANSWER
9 + 400 ÷ x

GRADING NOTE Form required: a constant plus a fraction.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Assume $x \neq -1$. Write $(x^2 + 3x + 5) \div (x + 1)$ in the form $q(x) + r \div (x + 1)$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

1515

QUESTION ID
S49-F1-Q2

TYPE
Short answer

CORRECT ANSWER
x + 2 + 3 ÷ (x + 1)

GRADING NOTE Form required: $q(x) + r \div (x + 1)$. $x^2 + 3x + 5 = (x + 1)(x + 2) + 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q2/5

Assume $x \neq 0$. Which expression equals $(6x^2 + 4x) \div (2x)$?

A. $3x + 4x$

B. $3x^2 + 2$

C. $6x + 2$

D. $3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

1516

QUESTION ID
S49-F1-Q3

TYPE
Multiple choice

CORRECT ANSWER
D. 3x + 2

GRADING NOTE Divide each term by 2x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F1-Q3**
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q3/5

True or false?

$(x^2 - 1) \div (x - 1) = x + 1$ for $x \neq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

1517

QUESTION ID
S49-F1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE $x^2 - 1 = (x - 1)(x + 1)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q4/5

True or false?

$(x + 8) \div x = 9$ for $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

1518

QUESTION ID
S49-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $(x + 8) \div x = 1 + 8 \div x$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5
High School — Algebra II (course placement in this bank) • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.6 • Set 49 • FORWARD 1 • Q5/5

SET 49 • HSF-BF.A.1.b

FORWARD 2

Graph and interpret an average-cost rational function • High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d

$A(x) = 9 + 400 \div x$ is the average cost per item, in dollars, for x items.

Give its horizontal asymptote and what it means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

1520

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

y = 9: for very many items, the average cost gets close to $9.

GRADING NOTE Form required: the asymptote and its meaning.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q1**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find A(100).

A(100) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

1521

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13

GRADING NOTE Any equivalent form. 9 + 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q2/5

What is the vertical asymptote of y = 400 ÷ x + 9?

A. x = 400

B. x = 9

C. y = 9

D. x = 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

1522

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x = 0

GRADING NOTE The function is undefined at x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q3/5

True or false?

As x increases, 9 + 400 ÷ x decreases toward 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

1523

QUESTION ID
S49-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE 400 ÷ x shrinks toward 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 9 + 400 \div x$ crosses the line $y = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

1524

QUESTION ID
S49-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE 400 ÷ x is never 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 49 • FORWARD 2 • Q5/5

Write arithmetic and geometric sequences recursively and with an explicit formula

MAIN	Algebra I main question	**HSF-BF.A.2**
BACKWARD 1	Evaluate recursively defined sequences	**HSF-IF.A.3**
BACKWARD 2	Generate a pattern from a rule and describe its features	**4.OA.C.5**
FORWARD 1	Sum geometric sequences	**HSA-SSE.B.4**
FORWARD 2	Use logarithms to find a term number	**HSF-LE.A.4**

SET 50 • HSF-BF.A.2

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.A.2

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula and a recursive formula for 7, 4, 1, −2, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

1527

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

a(n) = 10 − 3n;
a(1) = 7 and
a(n) = a(n − 1) − 3

GRADING NOTE Form required: both formulas. The explicit formula may be written 7 − 3(n − 1).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write an explicit formula and a recursive formula for 3, 12, 48, 192, ...

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

1528

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

$a(n) = 3 \cdot 4^{n-1}$;
$a(1) = 3$ and
$a(n) = 4a(n - 1)$

GRADING NOTE Form required: both formulas.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q2/5

An arithmetic sequence has a(3) = 14 and a(8) = 39.

Find a(1) and the common difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

1529

QUESTION ID

S50-M-Q3

TYPE

Short answer

CORRECT ANSWER

a(1) = 4; d = 5

GRADING NOTE Form required: both. 5 steps add 25, so d = 5, and a(1) = 14 − 2(5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q3/5

A geometric sequence of real numbers has a(1) = 2 and a(4) = 54.

What is the common ratio?

A. 3

B. 18

C. 27

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

1530

QUESTION ID

S50-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

GRADING NOTE $2r^3 = 54$, so $r^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q4/5

True or false?

The sequence 2, 4, 8, 14, ... is arithmetic.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

1531

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The differences are 2, 4, 6, which are not constant.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.A.2 • Set 50 • MAIN • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 1

Evaluate recursively defined sequences • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 4$ and $f(n) = f(n - 1) + 6$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

22

GRADING NOTE Any equivalent form. 4, 10, 16, 22.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(1) = 1$ and $f(n) = 2f(n - 1) + 1$.

Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Any equivalent form. 1, 3, 7, 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(n) = 2n - 1$.

What is $f(10)$?

A. 20

B. 21

C. 19

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 19

GRADING NOTE $2(10) - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A recursive formula gives each term from the term before it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

1536

QUESTION ID
S50-B1-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE That is what recursive means.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q4/5

True or false?

f(1) = 3 and f(n) = 2f(n − 1) give 3, 5, 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They give 3, 6, 12.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

BACKWARD 2

Generate a pattern from a rule and describe its features • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 5 and add 5.

Write the first 6 terms.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5, 10, 15, 20, 25, 30

GRADING NOTE Form required: the six terms in order.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Rule: start at 1 and add 4.

Explain why every term is odd.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

The first term is odd, and adding 4 (an even number) to an odd number gives an odd number.

GRADING NOTE Any equivalent explanation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q2/5

Rule: start at 2 and multiply by 2.

What is the 5th term?

A. 10

B. 16

C. 64

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 32

GRADING NOTE 2, 4, 8, 16, 32.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q3/5

True or false?

With the rule "start at 3 and add 3," every term is a multiple of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE 3, 6, 9, 12, ...

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q4/5

True or false?

With the rule "start at 10 and add 10," the 4th term is 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 10, 20, 30, 40.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 1

Sum geometric sequences • High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

ANSWER KEY Reduced copy of the question page, with its answer

Find the sum of the first 6 terms of 3, 12, 48, ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

1545

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4,095

GRADING NOTE Any equivalent form. $3(4^6 - 1) \div 3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q1/5

A job pays $1 on day 1, and the pay doubles each day.

What is the total pay for 10 days?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

1546

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$1,023

GRADING NOTE Any equivalent form. $2^{10} - 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q2**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

What is 5 + 10 + 20 + 40 + 80?

A. 160

B. 80

C. 155

D. 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

1547

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 155

GRADING NOTE $5(2^5 - 1) = 155$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F1-Q3**

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q3/5

True or false?

The formula $a(1 - r^n) \div (1 - r)$ works for $r = 0.5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

1548

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE It works for every $r \neq 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$2 + 6 + 18 + 54 = 2(3^4 - 1)$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

1549

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The sum is $80 = 2(3^4 - 1) \div 2$, not 160.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School — Algebra II (course placement in this bank) • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 50 • FORWARD 1 • Q5/5

SET 50 • HSF-BF.A.2

FORWARD 2

Use logarithms to find a term number • High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

ANSWER KEY Reduced copy of the question page, with its answer

$a(n) = 3 \cdot 4^{n-1}$.

What is the smallest n for which a(n) > 10,000?

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

1551

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $4^{n-1} > 3{,}333.3...$ gives $n - 1 > \log 3{,}333.3 \div \log 4 \approx 5.85$. Check: a(6) = 3,072 and a(7) = 12,288.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q1**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $2^{n-1} = 64$.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

1552

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

7

GRADING NOTE Any equivalent form. $64 = 2^6$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve $5 \cdot 2^n = 640$.

A. n = 6

B. n = 8

C. n = 7

D. n = 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

1553

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. n = 7

GRADING NOTE $2^n = 128 = 2^7$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q3**

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q3/5

True or false?

If $4^k = 64$, then $k = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

1554

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $4^3 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

If $3^n = 30$, then n = 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

1555

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3^{10} is much larger than 30; n = log 30 ÷ log 3 ≈ 3.10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S50-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 50 • FORWARD 2 • Q5/5

Identify the effect of replacing f(x) by f(x) + k, kf(x), f(kx), and f(x + k), and recognize even and odd functions

MAIN	Algebra I main question	**HSF-BF.B.3**
BACKWARD 1	Graph absolute value and square root functions	**HSF-IF.C.7.b**
BACKWARD 2	Read key features of quadratic graphs	**HSF-IF.C.7.a**
FORWARD 1	Describe a sequence of rigid transformations	**HSG-CO.A.5**
FORWARD 2	Transform sine and cosine graphs	**HSF-TF.B.5**

SET 51 • HSF-BF.B.3

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

$g(x) = (x - 4)^2 + 1$.

Describe how to get the graph of g from the graph of $y = x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

1558

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

Shift it 4 units right and 1 unit up.

GRADING NOTE Any equivalent description of both shifts.

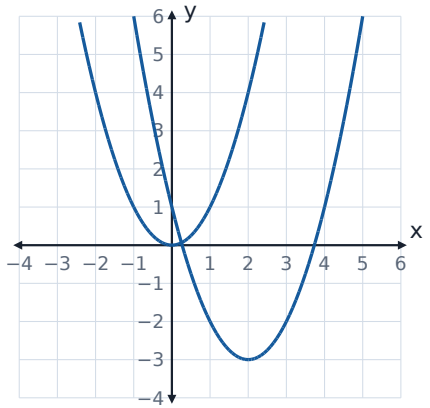
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The graph shows $y = x^2$ and a translated graph $y = g(x)$.

Write g(x).



g(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

1559

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 - 3$

GRADING NOTE Any equivalent form, such as $x^2 - 4x + 1$. The vertex moved from (0, 0) to (2, −3).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q2/5

$f(x) = x^2$. Compare $g(x) = 3f(x)$ and $h(x) = f(3x)$.

Which is a vertical stretch and which is a horizontal compression? Are g and h the same function?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

1560

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

$g(x) = 3x^2$ is a vertical stretch by 3; $h(x) = 9x^2$ is a horizontal compression by a factor of $\frac{1}{3}$. They are not the same.

GRADING NOTE Form required: both descriptions and the comparison. $(3x)^2 = 9x^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q3/5

$f(x) = x^3$.

Is f even, odd, or neither?

A. Odd: $f(-x) = -f(x)$

B. Even: $f(-x) = f(x)$

C. Both even and odd

D. Neither

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

1561

QUESTION ID

S51-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Odd: $f(-x) = -f(x)$

GRADING NOTE $(-x)^3 = -x^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q4/5

True or false?

The graph of y = f(x + 2) is the graph of y = f(x) shifted 2 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

1562

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is shifted 2 units left: the output at x is f's output at x + 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 51 • MAIN • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 1

Graph absolute value and square root functions • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

What is the vertex of $y = |x + 3| - 2$?

Vertex: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −2)

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q1/5

Give two points on the graph of $y = \sqrt{(x - 1)}$.

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

For example (1, 0) and (5, 2)

GRADING NOTE Any two points with $y = \sqrt{(x - 1)}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q2/5

What is the vertex of $y = -|x - 1| + 4$?

A. (1, 4)

B. (1, −4)

C. (4, 1)

D. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 4)

GRADING NOTE The expression inside is 0 at x = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q3/5

True or false?

y = |x| has its minimum at (0, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE |x| ≥ 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q4/5

True or false?

The domain of $y = \sqrt{(x + 2)}$ is $x \geq 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $x + 2 \geq 0$ gives $x \geq -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 51 • BACKWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

BACKWARD 2

Read key features of quadratic graphs • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a

Give the vertex of $y = (x - 2)^2 - 3$, and say whether it is a maximum or a minimum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(2, −3), a minimum

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-B2-Q1**

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the x-intercepts of $y = x^2 - 9$.

x-intercepts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 0) and (−3, 0)

GRADING NOTE Form required: both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q2/5

What is the y-intercept of $y = 2x^2 - 5x + 7$?

A. 7

B. 0

C. −5

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 7

GRADING NOTE Substitute x = 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q3/5

True or false?

The graph of $y = -x^2 + 4$ opens downward.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE The coefficient of x^2 is negative.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q4/5

True or false?

The vertex of $y = x^2 + 6$ is (6, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The vertex is (0, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.C.7.a • Set 51 • BACKWARD 2 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 1

Describe a sequence of rigid transformations • High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5

Describe a transformation that maps the triangle with vertices (0, 0), (2, 0), (0, 1) to the triangle with vertices (3, 2), (5, 2), (3, 3).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

1576

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A translation by (x, y) → (x + 3, y + 2).

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q1/5

The point (2, 5) is reflected across the x-axis and then translated 4 units left.

Where does it end up?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

1577

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, −5)

GRADING NOTE Any equivalent form. (2, 5) → (2, −5) → (−2, −5).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F1-Q2**

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q2/5

Which single transformation maps every (x, y) to (−x, −y)?

A. A translation

B. A rotation of 180° about the origin

C. A reflection across the y-axis

D. A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

1578

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A rotation of 180° about the origin

GRADING NOTE Both coordinates change sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q3/5

True or false?

Reflecting across the x-axis and then across the y-axis has the same effect as a 180° rotation about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

1579

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $(x, y) \to (x, -y) \to (-x, -y)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q4/5

True or false?

Translating and then reflecting always gives the same result as reflecting and then translating.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

1580

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For (1, 0), translating right 1 then reflecting across the y-axis gives (−2, 0); reflecting first then translating gives (0, 0).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School — Geometry (course placement in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 51 • FORWARD 1 • Q5/5

SET 51 • HSF-BF.B.3

FORWARD 2

Transform sine and cosine graphs • High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5

ANSWER KEY Reduced copy of the question page, with its answer

How is the graph of y = sin x + 3 related to the graph of y = sin x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

1582

QUESTION ID
S51-F2-Q1

TYPE
Short answer

CORRECT ANSWER
It is shifted 3 units up; its midline is y = 3.

GRADING NOTE Any equivalent description.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write a sine function with amplitude 2, midline y = 1, and period 2π.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

1583

QUESTION ID
S51-F2-Q2

TYPE
Short answer

CORRECT ANSWER
2sin x + 1

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q2**

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q2/5

How is y = 4sin x obtained from y = sin x?

A. A shift 4 units right

B. A horizontal stretch by a factor of 4

C. A vertical stretch by a factor of 4

D. A shift 4 units up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

1584

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A vertical stretch by a factor of 4

GRADING NOTE Every output is multiplied by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$y = -\sin x$ is the reflection of $y = \sin x$ across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

1585

QUESTION ID
S51-F2-Q4

TYPE
True or false

CORRECT ANSWER
A. True

GRADING NOTE Every output changes sign.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q4/5

True or false?

$y = \sin x - 2$ has midline $y = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

1586

QUESTION ID
S51-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Its midline is $y = -2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School — Algebra II (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 51 • FORWARD 2 • Q5/5

Solve f(x) = c for a simple function f and write an expression for the inverse

MAIN	Algebra I main question	**HSF-BF.B.4.a**
BACKWARD 1	Rearrange a formula for another variable	**HSA-CED.A.4**
BACKWARD 2	Use cube roots to solve $x^3 = p$	**8.EE.A.2**
FORWARD 1	Verify inverses by composition	**HSF-BF.B.4.b**
FORWARD 2	Restrict a domain to produce an inverse	**HSF-BF.B.4.d**

SET 52 • HSF-BF.B.4.a

MAIN

Algebra I main question • High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 2x^3$.

Solve $f(x) = 54$, and write an expression for the inverse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

1589

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 3$;

$f^{-1}(x) = \sqrt[3]{(x \div 2)}$

GRADING NOTE Form required: the solution and the inverse (any equivalent expression). $x^3 = 27$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = (x + 1) \div (x - 1)$, for $x \neq 1$.

Write an expression for the inverse.

$f^{-1}(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

1590

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

$(x + 1) \div (x - 1)$, for $x \neq 1$

GRADING NOTE Any equivalent form. Solve $y = (x + 1) \div (x - 1)$ for x: $x(y - 1) = y + 1$. This function is its own inverse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q2/5

F(C) = 1.8C + 32 gives degrees Fahrenheit.

Write the inverse C(F), and find C when F = 77.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

1591

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

C(F) = (F − 32) ÷ 1.8; 25 °C

GRADING NOTE Form required: the inverse (any equivalent form) and the value.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

$f(x) = 4x - 7$.

What is $f^{-1}(x)$?

A. $(x - 7) \div 4$

B. $(x + 7) \div 4$

C. $x \div 4 + 7$

D. $4x + 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

1592

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (x + 7) ÷ 4

GRADING NOTE Solve $y = 4x - 7$ for x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q4/5

True or false?

If $f(x) = x \div 3 + 2$, then $f^{-1}(x) = 3x - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

1593

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE $x = y \div 3 + 2$ gives $y = 3(x - 2)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.4.a • Set 52 • MAIN • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 1

Rearrange a formula for another variable • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Solve y = 4x − 7 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(y + 7) ÷ 4

GRADING NOTE Any equivalent form.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve P = 2l + 2w for l.

l = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(P − 2w) ÷ 2

GRADING NOTE Any equivalent form, such as P ÷ 2 − w.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve d = rt for t. (r ≠ 0)

A. t = r ÷ d

B. t = d − r

C. t = d ÷ r

D. t = dr

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. t = d ÷ r

GRADING NOTE Divide both sides by r.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q3/5

True or false?

Solving $y = x^3$ for x gives $x = \sqrt[3]{y}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

GRADING NOTE Take the cube root of both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Solving y = 5x for x gives x = 5y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = y ÷ 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • HSF-BF.B.4.a

BACKWARD 2

Use cube roots to solve $x^3 = p$ • Grade 8 • Expressions & Equations • 8.EE.A.2